Computer Architecture — Complete Flashcards Companion

Comprehensive Exam Preparation & Conceptual Q&A; Review across Chapters 4 & 5

Chapter 4.5 - Pipelining Overview & Hazards

#1	**Q1: What is pipelining in the context of computer architecture?**
	Answer: It is an implementation technique in which multiple instructions are overlapped in execution, much like an assembly line.

#2	**Q2: What is the primary performance goal of pipelining?**
	Answer: Pipelining aims to improve instruction throughput, which is the number of instructions completed per unit of time.

#3	**Q3: What is the 'pipelining paradox' regarding instruction execution time?**
	Answer: The time to complete a single instruction (latency) is not shorter, but the total time for many instructions is shorter due to increased throughput.

#4	**Q4: Under ideal conditions with a large number of instructions, the speed-up from pipelining is approximately equal to what?**
	Answer: The number of pipe stages.

#5	**Q5: What are the five classic stages of a RISC-V instruction pipeline?**
	Answer: 1. Fetch instruction, 2. Decode instruction/Read registers, 3. Execute, 4. Access memory, 5. Write back result.

#6	**Q6: In a pipelined processor, what determines the duration of the clock cycle?**
	Answer: The duration of the slowest pipeline stage.

#7	**Q7: In a non-pipelined, single-cycle processor, what determines the duration of the clock cycle?**
	Answer: The time required to execute the slowest possible instruction.

#8	**Q8: What is a pipeline hazard?**
	Answer: A situation in pipelining when the next instruction cannot execute in the following clock cycle for some reason.

#9	**Q9: Events in a pipeline such as start-up and _____ affect performance when the number of tasks is not large compared to the number of stages.**
	Answer: wind-down

#10	**Q10: What is a structural hazard in a pipeline?**
	Answer: A situation where a planned instruction cannot execute because the hardware cannot support the combination of instructions set to execute in the same clock cycle.

#11	**Q11: What is a data hazard in a pipeline?**
	Answer: A situation where an instruction cannot execute in the proper clock cycle because data it needs is not yet available from a previous instruction.

#12	**Q12: What is a control hazard in a pipeline?**
	Answer: A situation where the proper instruction cannot execute because the instruction that was fetched is not the one that is needed, typically due to a branch.

#13	**Q13: What is another common name for a control hazard?**
	Answer: A branch hazard.

#14	**Q14: How does having a fixed length for all RISC-V instructions simplify pipelining?**
	Answer: It makes it much easier to fetch instructions in the first pipeline stage and decode them in the second stage.

#15	**Q15: How does RISC-V being a load-store architecture simplify pipelining?**
	Answer: It allows the memory address calculation and memory access to occur in predictable, separate stages (EX and MEM).

#16	**Q16: A specific type of data hazard where data from a \`load\` instruction is needed by a subsequent instruction before it's available is called a _____.**
	Answer: load-use data hazard

#17	**Q17: What is forwarding, also known as bypassing, used for?**
	Answer: It is a method of resolving a data hazard by retrieving the missing data from internal buffers rather than waiting for it to be written to the register file.

#18	**Q18: Why is forwarding sometimes insufficient, such as in a load-use data hazard?**
	Answer: The desired data is not available until after the memory access stage, which is too late for the execution stage of the immediately following instruction.

#19	**Q19: A stall initiated in order to resolve a hazard is often called a _____.**
	Answer: bubble

#20	**Q20: Besides hardware stalls, what is a software technique a compiler can use to mitigate data hazards?**
	Answer: Reordering code to separate a dependent instruction from the one that produces its data.

#21	**Q21: What is the simplest, albeit slow, hardware solution to a control hazard caused by a branch?**
	Answer: To stall the pipeline after fetching a branch until the branch outcome and target address are known.

#22	**Q22: A method of resolving a branch hazard by assuming an outcome and proceeding rather than waiting is known as _____.**
	Answer: branch prediction

#23	**Q23: What is a simple static branch prediction strategy mentioned in the text?**
	Answer: To predict that conditional branches will always be untaken.

#24	**Q24: What is dynamic branch prediction?**
	Answer: A technique where hardware makes predictions based on the past behavior of each individual conditional branch, and may change predictions during program execution.

#25	**Q25: What must the pipeline control do when a branch prediction is incorrect?**
	Answer: It must ensure the wrongly fetched instructions have no effect and restart the pipeline from the correct branch address.

#26	**Q26: An architectural solution to control hazards, used by MIPS, that always executes the instruction immediately following a branch is called a _____.**
	Answer: delayed branch

#27	**Q27: Pipelining improves instruction _____ rather than instruction ____.**
	Answer: throughput, latency

#28	**Q28: What is pipeline latency?**
	Answer: The number of stages in a pipeline, representing the time it takes for a single instruction to complete its execution.

#29	**Q29: In the laundry analogy, if the washer, dryer, folder, and storer each take 30 minutes, how long does pipelined laundry take for four loads?**
	Answer: 3.5 hours.

#30	**Q30: In the laundry analogy, if the washer, dryer, folder, and storer each take 30 minutes, how long does sequential laundry take for four loads?**
	Answer: 8 hours.

#31	**Q31: What are the five steps of instruction execution for RISC-V, which form the basis for its five-stage pipeline?**
	Answer: 1. Fetch instruction, 2. Read registers/decode, 3. Execute/calculate address, 4. Access memory operand, 5. Write result to register.

#32	**Q32: In the RISC-V single-cycle example, the clock cycle was 800 ps because it had to be stretched to accommodate the slowest instruction, which was ____.**
	Answer: ld (load doubleword)

#33	**Q33: In the graphical representation of forwarding in Figure 4.27, the path goes from the output of the EX stage of the \`add\` instruction to the input of the EX stage of which instruction?**
	Answer: The \`sub\` instruction.

#34	**Q34: In the code reordering example, which instruction was moved up to eliminate the pipeline stalls?**
	Answer: The third \`ld\` instruction (\`ld x4, 16(x31)\`).

#35	**Q35: Why is it impossible for a forwarding path to go from the MEM stage of one instruction to the EX stage of the *following* instruction without a stall?**
	Answer: This would mean going backward in time, as the following instruction's EX stage occurs at the same time as the current instruction's MEM stage.

#36	**Q36: Assuming conditional branches are 17% of instructions, what is the CPI if every branch causes a one-cycle stall and all other instructions have a CPI of 1?**
	Answer: The CPI would be 1.17.

#37	**Q37: In a pipeline diagram, what does shading on the right half of the register file symbol indicate?**
	Answer: It indicates that the register file is being read in that stage.

#38	**Q38: In a pipeline diagram, what does shading on the left half of the register file symbol indicate?**
	Answer: It indicates that the register file is being written in that stage.

#39	**Q39: The text states that modern implementations of the x86 architecture often translate complex x86 instructions into simpler, RISC-V-like operations before pipelining them. Why is this done?**
	Answer: Because x86 instructions vary in length from 1 to 15 bytes, making direct pipelining considerably more challenging.

#40	**Q40: In what pipeline stage does a RISC-V instruction write its result?**
	Answer: The last stage (Write-back or WB).

#41	**Q41: What is the primary performance metric improved by pipelining?**
	Answer: Pipelining improves instruction throughput, which is the rate at which instructions are completed.

#42	**Q42: What is the 'pipelining paradox'?**
	Answer: The time to complete a single instruction (latency) is not shorter, but the total time for many instructions is shorter due to improved throughput.

#43	**Q43: In the laundry analogy, what are the four sequential stages of a non-pipelined approach?**
	Answer: 1. Wash, 2. Dry, 3. Fold, 4. Put away.

#44	**Q44: In the laundry analogy, at what point are all pipeline stages operating concurrently?**
	Answer: When the fourth load is in the washer, the third is in the dryer, the second is being folded, and the first is being put away.

#45	**Q45: Under ideal conditions, the speed-up from pipelining is approximately equal to the _____.**
	Answer: number of pipe stages

#46	**Q46: Why is the speed-up in the laundry example (Figure 4.23) only 2.3 times instead of the potential 4 times?**
	Answer: Because the pipeline is not completely full during the start-up and wind-down phases with only four loads of laundry.

#47	**Q47: In a single-cycle implementation, what determines the length of the clock cycle?**
	Answer: The clock cycle must be stretched to accommodate the slowest instruction.

#48	**Q48: In a pipelined implementation, what determines the length of the clock cycle?**
	Answer: The clock cycle must be long enough to accommodate the slowest pipeline stage.

#49	**Q49: In the example from Figure 4.24, what was the single-cycle clock time and what was the pipelined clock time?**
	Answer: The single-cycle time was 800 ps (for the slowest instruction, ld), and the pipelined time was 200 ps (for the slowest stage, memory access or ALU).

#50	**Q50: Pipelining improves performance by increasing instruction _____, in contrast to decreasing the _____ of an individual instruction.**
	Answer: throughput; execution time (or latency)

#51	**Q51: What is one feature of the RISC-V instruction set that makes it easier to pipeline?**
	Answer: All RISC-V instructions are the same length, which simplifies fetching and decoding.

#52	**Q52: What is a second feature of the RISC-V instruction set design that aids pipelining?**
	Answer: It has only a few instruction formats, with source/destination register fields in the same location.

#53	**Q53: What is a third feature of the RISC-V instruction set that simplifies pipelining?**
	Answer: Memory operands only appear in load or store instructions, allowing address calculation in one stage and memory access in the next.

#54	**Q54: Term: Structural Hazard**
	Answer: A hazard that occurs when the hardware cannot support the combination of instructions set to execute in the same clock cycle.

#55	**Q55: What is a classic example of a structural hazard in a processor pipeline?**
	Answer: When one instruction is accessing data memory while another instruction is being fetched from memory, if there is only a single memory unit.

#56	**Q56: Term: Data Hazard**
	Answer: A hazard that occurs when a planned instruction cannot execute because data it needs is not yet available.

#57	**Q57: What causes a data hazard in a computer pipeline?**
	Answer: It arises from the dependence of one instruction on an earlier one that is still in the pipeline.

#58	**Q58: What is the primary hardware-based solution for resolving many data hazards?**
	Answer: Forwarding (or bypassing), which retrieves the missing data from internal buffers rather than waiting for it to be written to a register.

#59	**Q59: In the sequence \`add x19, x0, x1\` followed by \`sub x2, x19, x3\`, how does forwarding resolve the data hazard?**
	Answer: The result from the ALU in the EX stage of the \`add\` instruction is forwarded directly to the ALU input for the EX stage of the \`sub\` instruction.

#60	**Q60: A forwarding path is only valid if the destination stage is _____ in time than the source stage.**
	Answer: later

#61	**Q61: Term: Load-use data hazard**
	Answer: A specific data hazard where data being loaded has not yet become available when needed by a subsequent instruction.
#62	**Q62: Why can't forwarding alone resolve a load-use data hazard without a stall?**
	Answer: The desired data is only available after the memory access (MEM) stage, which is too late for the execution (EX) stage of the immediately following instruction.
#63	**Q63: What is a pipeline stall, also known as a bubble?**
	Answer: A stall initiated in order to resolve a hazard, effectively inserting a delay into the pipeline.
#64	**Q64: In the example `a = b + e; c = b + f;`, how can the code be reordered to avoid pipeline stalls?**
	Answer: By moving the load for `f` up, between the loads for `b` and `e` and the first `add` instruction, to separate the dependent instructions.
#65	**Q65: Term: Control Hazard**
	Answer: Also called a branch hazard, it occurs when the proper instruction cannot execute because the instruction that was fetched is not the one that is needed.
#66	**Q66: What is the simplest, though slow, solution to a control hazard caused by a conditional branch?**
	Answer: To stall the pipeline after fetching a branch until the outcome is determined.
#67	**Q67: Assuming a 1-cycle stall for every conditional branch, what would be the impact on CPI if branches are 17% of instructions?**
	Answer: The CPI would increase from an ideal 1.0 to 1.17, representing a 17% slowdown.
#68	**Q68: What is branch prediction?**
	Answer: A method of resolving a branch hazard that assumes a given outcome and proceeds from that assumption rather than waiting.
#69	**Q69: What happens in a 'predict-not-taken' scheme if the branch is actually taken?**
	Answer: The pipeline stalls, and the incorrectly fetched instructions are flushed while the pipeline restarts from the correct branch address.
#70	**Q70: What is the key difference between static and dynamic branch prediction?**
	Answer: Static prediction is rigid (e.g., always predict taken for backward branches), while dynamic prediction uses the past behavior of each specific branch to make its guess.

#71	**Q71: What is the typical accuracy of modern dynamic branch predictors?**
	Answer: They can correctly predict conditional branches with more than 90% accuracy.

#72	**Q72: What is a 'delayed branch'?**
	Answer: A solution to control hazards where the instruction immediately following the branch is always executed, regardless of the branch outcome.

#73	**Q73: For the code sequence: `ld x10, 0(x10)` followed by `add x11, x10, x10`, can it execute without stalling?**
	Answer: No, this is a load-use hazard that must stall for one cycle, even with forwarding.

#74	**Q74: For the code sequence: `add x11, x10, x10` followed by `addi x12, x11, 5`, can it execute without stalling?**
	Answer: Yes, this data hazard can be resolved using only forwarding.

#75	**Q75: For the code sequence: `addi x11, x10, 1` followed by `addi x12, x10, 2` and `addi x13, x10, 3`, can it execute without issues?**
	Answer: Yes, it can execute without stalling or forwarding because there are no data dependencies between the instructions.

#76	**Q76: What is the definition of pipeline latency?**
	Answer: The number of stages in a pipeline or the number of stages between two instructions during execution.

Chapter 4.6 - Pipelined Datapath & Control

#77	**Q77: What is the first stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The first stage is Instruction Fetch, abbreviated as IF.

#78	**Q78: What is the second stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The second stage is Instruction Decode and register file read, abbreviated as ID.

#79	**Q79: What is the third stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The third stage is Execution or address calculation, abbreviated as EX.

#80	**Q80: What is the fourth stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The fourth stage is Data memory access, abbreviated as MEM.

#81	**Q81: What is the fifth stage of the five-stage pipeline, and what is its abbreviation?**
	Answer: The fifth stage is Write back, abbreviated as WB.

#82	**Q82: In a five-stage pipeline, up to how many instructions can be in execution during a single clock cycle?**
	Answer: Up to five instructions can be in execution during any single clock cycle.

#83	**Q83: What is the general direction of flow for instructions and data through the pipeline stages?**
	Answer: Instructions and data generally move from left to right through the five stages.

#84	**Q84: What is the first exception to the left-to-right flow of data in the pipelined datapath?**
	Answer: The write-back stage, which sends the result back to the register file in the middle of the datapath.

#85	**Q85: What is the second exception to the left-to-right flow of data in the pipelined datapath?**
	Answer: The selection of the next PC value, which chooses between the incremented PC and a branch address from the MEM stage.

#86	**Q86: The right-to-left data flow of the write-back stage can lead to what type of hazard?**
	Answer: This flow can lead to data hazards.

#87	**Q87: The right-to-left data flow for selecting the next PC value can lead to what type of hazard?**
	Answer: This flow can lead to control hazards.

#88	**Q88: To share a single datapath among multiple instructions in a pipeline, what hardware components must be added between stages?**
	Answer: Pipeline registers must be added to hold data between stages.

#89	**Q89: What is the naming convention for pipeline registers?**
	Answer: They are named for the two stages they separate (e.g., the register between IF and ID is named IF/ID).

#90	**Q90: Why is there no pipeline register at the end of the write-back (WB) stage?**
	Answer: It is redundant because all instructions must update some processor state (register file, memory, or PC) which already holds the result.

#91	**Q91: Which component of the processor's visible architectural state can be thought of as a pipeline register that feeds the IF stage?**
	Answer: The Program Counter (PC) can be thought of as a pipeline register feeding the IF stage.

#92	**Q92: What two pieces of information are placed into the IF/ID pipeline register during the Instruction Fetch stage?**
	Answer: The fetched instruction and the incremented PC address are placed in the IF/ID register.

#93	**Q93: During the ID stage of a load instruction, what values are stored in the ID/EX pipeline register?**
	Answer: The two read register values, the sign-extended immediate value, and the PC address are stored in the ID/EX register.

#94	**Q94: What calculation is performed by the ALU during the EX stage of a load (`ld`) instruction?**
	Answer: The ALU adds the contents of a register and the sign-extended immediate to calculate the effective memory address.

#95	**Q95: During the MEM stage of a load (`ld`) instruction, data is read from memory and placed into which pipeline register?**
	Answer: The data read from memory is placed into the MEM/WB pipeline register.

#96	**Q96: What is the final action performed during the WB stage of a load (`ld`) instruction?**
	Answer: Data is read from the MEM/WB pipeline register and written into the register file.

#97	**Q97: For a store instruction, what is passed from the ID/EX pipeline register to the EX/MEM pipeline register to be used in the MEM stage?**
	Answer: The register data to be stored is passed into the EX/MEM pipeline register.

#98	**Q98: What is the primary action during the MEM stage of a store (`sd`) instruction?**
	Answer: The data from the EX/MEM pipeline register is written into data memory.

#99	**Q99: What happens during the write-back (WB) stage for a store (`sd`) instruction?**
	Answer: Nothing happens in the write-back stage for a store instruction.

#100	**Q100: To prevent a structural hazard, each logical component of the datapath (e.g., ALU, data memory) can be used within how many pipeline stages?**
	Answer: Each logical component can be used only within a single pipeline stage.

#101	**Q101: What was the design bug in the initial pipelined datapath concerning the load (`ld`) instruction?**
	Answer: The write register number was not passed down the pipeline, so the wrong instruction would supply it during the WB stage.

#102	**Q102: How was the design bug for the load (`ld`) instruction fixed?**
	Answer: The destination register number is passed from the ID stage through the ID/EX, EX/MEM, and MEM/WB pipeline registers for use in the WB stage.

#103	**Q103: What is the key difference between a multiple-clock-cycle pipeline diagram and a single-clock-cycle pipeline diagram?**
	Answer: A multiple-clock-cycle diagram shows instruction execution over time, while a single-clock-cycle diagram shows the state of the entire datapath at one specific clock cycle.

#104	**Q104: A single-clock-cycle diagram represents a _____ slice of a multiple-clock-cycle diagram.**
	Answer: vertical

#105	**Q105: Why does allowing some instructions to take fewer cycles not necessarily improve pipeline throughput?**
	Answer: Throughput is determined by the clock cycle, not the latency (number of stages) of individual instructions.

#106	**Q106: Instead of making instructions take fewer cycles, what alternative approach could improve pipeline performance?**
	Answer: Making the pipeline longer so instructions take more, but shorter, clock cycles could improve performance.

#107	**Q107: Control signals for the EX, MEM, and WB stages are created during which pipeline stage?**
	Answer: The control signals are created during the instruction decode (ID) stage.

#108	**Q108: How are control signals transported to the correct pipeline stage for execution?**
	Answer: They are passed down the pipeline by extending the pipeline registers (ID/EX, EX/MEM, MEM/WB) to include control bits.

#109	**Q109: Which two control lines are set in the Execution/address calculation (EX) stage?**
	Answer: The ALUOp and ALUSrc control lines are set in the EX stage.

#110	**Q110: Which three control lines are set in the Memory access (MEM) stage?**
	Answer: The Branch, MemRead, and MemWrite control lines are set in the MEM stage.

#11 1	**Q111: Which two control lines are set in the Write-back (WB) stage?**
	Answer: The MemtoReg and RegWrite control lines are set in the WB stage.

#11 2	**Q112: The control signals created in the ID stage are first placed into which pipeline register?**
	Answer: They are placed into the ID/EX pipeline register.

#11 3	**Q113: Why is an instruction passed through a pipeline stage even if there is nothing for it to do in that stage?**
	Answer: This is done because later instructions are already progressing at the maximum rate, and there is no way to accelerate them.

#11 4	**Q114: What information must be preserved and passed down the pipeline for a load instruction to function correctly?**
	Answer: The destination register number must be preserved and passed through the pipeline registers to the WB stage.

#11 5	**Q115: In the stylized datapath diagrams, the register file is broken into two logical parts to represent its dual use in which two stages?**
	Answer: It is used for register reads during the ID stage and register writes during the WB stage.

#11 6	**Q116: What is the purpose of the ALUSrc control signal in the EX stage?**
	Answer: It selects whether the second ALU input is from a register (Read data 2) or the sign-extended immediate.

#11 7	**Q117: What is the purpose of the MemtoReg control signal in the WB stage?**
	Answer: It decides whether the value written to the register file comes from the ALU result or from data memory.

#11 8	**Q118: What are the five stages of a pipelined datapath in order?**
	Answer: 1. Instruction fetch (IF), 2. Instruction decode (ID), 3. Execution (EX), 4. Memory access (MEM), 5. Write back (WB).

#11 9	**Q119: What is the primary activity in the Instruction Fetch (IF) stage of a pipeline?**
	Answer: Reading an instruction from memory using the address in the Program Counter (PC).

#12 0	**Q120: What are the two main activities in the Instruction Decode (ID) stage of a pipeline?**
	Answer: Decoding the instruction and reading the contents of the specified registers from the register file.

#121	**Q121: What is the primary activity in the Execution (EX) stage of a pipeline?**
	Answer: Performing an operation in the ALU, such as an arithmetic calculation or an address calculation.

#122	**Q122: What is the primary activity in the Data Memory Access (MEM) stage of a pipeline?**
	Answer: Reading from or writing to the data memory.

#123	**Q123: What is the primary activity in the Write Back (WB) stage of a pipeline?**
	Answer: Writing the result of the instruction (from the ALU or memory) back into the register file.

#124	**Q124: In a five-stage pipeline, how many instructions can be in execution during a single clock cycle?**
	Answer: Up to five instructions.

#125	**Q125: What is the general direction of data and instruction flow through the pipelined datapath stages?**
	Answer: From left to right, from the IF stage to the WB stage.

#126	**Q126: What is the first exception to the left-to-right data flow in a pipelined datapath?**
	Answer: The write-back stage, which sends the result back to the register file located in the middle of the datapath.

#127	**Q127: What is the second exception to the left-to-right data flow in a pipelined datapath?**
	Answer: The selection of the next PC value, which can choose a branch address from the MEM stage.

#128	**Q128: The right-to-left data flow from the write-back stage can lead to what type of hazard?**
	Answer: Data hazards.

#129	**Q129: The right-to-left data flow for PC selection (branching) can lead to what type of hazard?**
	Answer: Control hazards.

#130	**Q130: What hardware components are added between the stages of a datapath to implement pipelining?**
	Answer: Pipeline registers.

#131	**Q131: The pipeline register located between the Instruction Fetch and Instruction Decode stages is named _____.**
	Answer: IF/ID.

#132	**Q132: What is the purpose of pipeline registers in a pipelined datapath?**
	Answer: To hold the data and control information for an instruction as it moves from one stage to the next, allowing stages to be shared by different instructions simultaneously.

#133	**Q133: Which component of the architectural state can be thought of as a pipeline register that feeds the IF stage?**
	Answer: The Program Counter (PC).

#134	**Q134: In the IF stage of a \`ld\` instruction, what happens to the PC value?**
	Answer: It is incremented by 4 and written back into the PC for the next cycle, and a copy is saved in the IF/ID pipeline register.

#135	**Q135: In the ID stage of a \`ld\` instruction, what values are stored in the ID/EX pipeline register?**
	Answer: The values from two read registers, the sign-extended immediate value, and the PC address.

#136	**Q136: What calculation is performed in the EX stage of a \`ld\` instruction?**
	Answer: The ALU adds the contents of a register to the sign-extended immediate value to calculate the memory address.

#137	**Q137: What happens in the MEM stage of a \`ld\` instruction?**
	Answer: The data memory is read using the address from the EX/MEM pipeline register, and the loaded data is placed in the MEM/WB pipeline register.

#138	**Q138: In the WB stage of a \`ld\` instruction, data is read from the _____ pipeline register and written into the register file.**
	Answer: MEM/WB

#139	**Q139: How does a \`store\` instruction make the data to be stored available during the MEM stage?**
	Answer: The data is read from the register file in the ID stage and passed through the ID/EX and EX/MEM pipeline registers.

#140	**Q140: What happens in the Write-Back (WB) stage for a `store` instruction?**
	Answer: Nothing happens in the WB stage for a store instruction, as the result was already written to memory in the MEM stage.

#141	**Q141: To avoid structural hazards, each logical component of the datapath, such as the ALU, can be used only within a single ____.**
	Answer: pipeline stage

#142	**Q142: What was the original design bug in handling the `ld` instruction's write-back stage?**
	Answer: The write register number was being supplied by the instruction in the IF/ID pipeline register, not the `ld` instruction that was actually in the WB stage.

#143	**Q143: How was the `ld` instruction bug fixed in the pipelined datapath design?**
	Answer: The destination register number is passed from the ID stage through the ID/EX, EX/MEM, and MEM/WB pipeline registers to be used in the WB stage.

#144	**Q144: What type of pipeline diagram shows the complete execution of multiple instructions over time, with clock cycles on one axis and instructions on the other?**
	Answer: A multiple-clock-cycle pipeline diagram.

#145	**Q145: What type of pipeline diagram shows the state of the entire datapath during a single clock cycle?**
	Answer: A single-clock-cycle pipeline diagram.

#146	**Q146: A single-clock-cycle diagram represents a vertical slice through what other type of diagram?**
	Answer: A multiple-clock-cycle diagram.

#147	**Q147: How are control signals for the EX, MEM, and WB stages passed down the pipeline?**
	Answer: They are generated in the ID stage and then stored in the control portions of the pipeline registers (ID/EX, EX/MEM, MEM/WB).

#148	**Q148: Which two control signals are set during the Execution/address calculation (EX) stage?**
	Answer: ALUOp and ALUSrc.

#149	**Q149: Which three control signals are set during the Memory access (MEM) stage?**
	Answer: Branch, MemRead, and MemWrite.

#150	**Q150: Which two control signals are set during the Write-back (WB) stage?**
	Answer: MemtoReg and RegWrite.

#151	**Q151: Why do the IF and ID stages not require special control signals to be passed to them?**
	Answer: Their primary operations (reading instruction memory, writing the PC, reading registers) are always asserted or are determined by fixed fields in the instruction format.

#152	**Q152: In the corrected datapath, the write register number for the WB stage comes from which pipeline register?**
	Answer: The MEM/WB pipeline register.

#153	**Q153: What information must be stored in the IF/ID pipeline register for a `beq` instruction to potentially work later on?**
	Answer: The incremented PC address.

#154	**Q154: During the EX stage of a `store` instruction, what two values are placed into the EX/MEM pipeline register?**
	Answer: The calculated effective address and the second register value (the data to be stored).

#155	**Q155: According to the 'Check Yourself' section, is it true that allowing some instructions to take fewer stages always increases pipeline performance?**
	Answer: No, because pipeline throughput is determined by the clock cycle time, not the latency of individual instructions.

#156	**Q156: The number of pipeline stages per instruction primarily affects an instruction's _____, not the pipeline's throughput.**
	Answer: latency

#157	**Q157: According to the 'Check Yourself' section, what is a potential benefit of making a pipeline longer (more stages)?**
	Answer: A longer pipeline can allow for a shorter clock cycle time, which could improve overall performance (throughput).

#158	**Q158: The IF/ID pipeline register must be wide enough to hold both the 32-bit instruction and the incremented ____ address.**
	Answer: 64-bit PC

#159	**Q159: The control logic generates control signals for the EX, MEM, and WB stages during which pipeline stage?**
	Answer: The Instruction Decode (ID) stage.

#160	**Q160: The control line _____ determines whether the ALU gets its second operand from the register file or the sign-extended immediate field.**
	Answer: ALUSrc

#161	**Q161: The control line _____ determines whether the value written to the register file comes from the ALU result or from data memory.**
	Answer: MemtoReg

#162	**Q162: The _____ control signal must be asserted to enable writing to the register file in the WB stage.**
	Answer: RegWrite

#163	**Q163: What two signals must both be asserted for the `PCSrc` multiplexor to select the branch target address?**
	Answer: The `Branch` signal (from control) and the `Zero` signal (from the ALU).

Chapter 4.8 - Control Hazards & Branch Prediction

#164	**Q164: What is a control hazard in a processor pipeline?**
	Answer: It is a delay in determining the proper instruction to fetch, caused by a conditional branch instruction.

#165	**Q165: In the non-optimized five-stage pipeline described, in which stage is the decision about whether to branch made?**
	Answer: The decision is made in the MEM (Memory Access) pipeline stage.

#166	**Q166: Without any optimization, how many sequential instructions following a branch are fetched before the branch decision is made in the MEM stage?**
	Answer: Three sequential instructions are fetched and begin execution.

#167	**Q167: The technique of discarding instructions in a pipeline, usually due to an unexpected event like a mispredicted branch, is called _____.**
	Answer: flushing

#168	**Q168: What is the simplest static branch prediction scheme mentioned in the text?**
	Answer: The 'Assume Branch Not Taken' scheme, where execution continues down the sequential instruction stream.

#169	**Q169: In the 'Assume Branch Not Taken' scheme, what happens if the conditional branch is actually taken?**
	Answer: The instructions that were being fetched and decoded must be discarded, and execution continues at the branch target.

#170	**Q170: How are instructions discarded or flushed when a branch is mispredicted in the 'Assume Not Taken' scheme?**
	Answer: The original control values for the instructions in the IF, ID, and EX stages are changed to 0s.

#171	**Q171: What is the primary method discussed to improve conditional branch performance by reducing the cost of a taken branch?**
	Answer: Moving the conditional branch execution earlier in the pipeline, from the MEM stage to the ID stage.

#172	**Q172: What are the two actions that must occur earlier to move conditional branch execution to the ID stage?**
	Answer: Computing the branch target address and evaluating the branch decision must both occur earlier.

#173	**Q173: What hardware modification is needed in the ID stage to calculate the branch target address earlier?**
	Answer: The branch adder is moved from the EX stage to the ID stage.

#174	**Q174: What hardware modification is needed in the ID stage to evaluate the branch decision for 'branch if equal' earlier?**
	Answer: An equality test unit is added to compare two register values during the ID stage.

#175	**Q175: When branch execution is moved to the ID stage, what new hardware complication arises regarding data dependencies?**
	Answer: New forwarding logic is required, as the equality test unit in ID may need results from later pipeline stages (EX/MEM or MEM/WB).

#176	**Q176: If an ALU instruction immediately precedes a branch that depends on its result, what is required when branch execution is in the ID stage?**
	Answer: A stall will be required because the ALU result is produced in the EX stage, after the branch's ID cycle.

#177	**Q177: If a load instruction is immediately followed by a conditional branch that depends on its result, how many stall cycles are needed if branch execution is in the ID stage?**
	Answer: Two stall cycles are needed, as the load result is not available until the end of the MEM cycle.

#178	**Q178: By moving branch execution to the ID stage, the penalty of a taken branch is reduced to how many instructions?**
	Answer: The penalty is reduced to only one instruction, which is the one currently being fetched.

#179	**Q179: What is the purpose of the `IF.Flush` control line?**
	Answer: It zeros the instruction field of the IF/ID pipeline register, effectively transforming the fetched instruction into a nop.

#180	**Q180: Prediction of branches at runtime using runtime information is known as _____.**
	Answer: dynamic branch prediction

#181	**Q181: What is a branch prediction buffer, also known as a branch history table?**
	Answer: It is a small memory indexed by the lower portion of a branch instruction's address, containing bits that indicate if the branch was recently taken.

#182	**Q182: What is the primary performance shortcoming of a simple 1-bit dynamic branch prediction scheme?**
	Answer: If a branch is almost always taken (like in a loop), it will be mispredicted twice for each full loop execution rather than just once.

#183	**Q183: For a loop branch that is taken nine times and not taken once, what is the prediction accuracy using a 1-bit scheme?**
	Answer: The accuracy is 80%, with two incorrect predictions (on the first and last iterations) and eight correct ones.

#184	**Q184: Why does a 1-bit predictor mispredict the first iteration of a loop that was just completed?**
	Answer: The prediction bit was flipped to 'not taken' on the final, exiting iteration of the previous loop execution.

#185	**Q185: How does a 2-bit branch prediction scheme improve over a 1-bit scheme?**
	Answer: A prediction must be wrong twice before the prediction state is changed, leading to only one misprediction for highly regular loops.

#186	**Q186: In a 2-bit counter-based predictor, what action is taken when the prediction is accurate?**
	Answer: The counter is incremented (or moved towards a more strongly-taken/not-taken state).

#187	**Q187: In a 2-bit counter-based predictor, what action is taken when the prediction is inaccurate?**
	Answer: The counter is decremented (or moved towards the weakly-taken/not-taken state, and eventually flips).

#188	**Q188: What structure caches the destination PC or destination instruction for a branch to reduce the taken-branch penalty?**
	Answer: A branch target buffer.

#189	**Q189: How does a branch target buffer differ from a simpler branch prediction buffer?**
	Answer: It is usually organized as a cache with tags, making it more costly, and it stores the target address or instruction, not just a prediction bit.

#190	**Q190: What is a correlating predictor?**
	Answer: A branch predictor that combines the local behavior of a branch with global information about recently executed branches.

#191	**Q191: What is a tournament branch predictor?**
	Answer: A predictor that uses multiple prediction schemes and a selection mechanism to choose the best one for a given branch.

#192	**Q192: What type of instruction can be added to an instruction set to reduce the number of conditional branches?**
	Answer: Conditional move instructions.

#193	**Q193: What does the ARMv8 `CSEL` instruction do?**
	Answer: It conditionally selects one of two source registers to copy to a destination register based on a condition code.

#194	**Q194: For a branch with 5% taken frequency and a 2-cycle misprediction penalty, which scheme is best: predict-taken, predict-not-taken, or 90% accurate dynamic?**
	Answer: The predict-not-taken scheme is best, as it will be correct 95% of the time.
#195	**Q195: For a branch with 95% taken frequency and a 2-cycle misprediction penalty, which scheme is best: predict-taken, predict-not-taken, or 90% accurate dynamic?**
	Answer: The predict-taken scheme is best, as it will be correct 95% of the time.
#196	**Q196: For a branch with 70% taken frequency and a 2-cycle misprediction penalty, which scheme is best: predict-taken, predict-not-taken, or 90% accurate dynamic?**
	Answer: The dynamic predictor is best, as its 90% accuracy is higher than predict-taken (70%) or predict-not-taken (30%).
#197	**Q197: What is the name for a pipeline hazard involving conditional branches?**
	Answer: It is called a control hazard or a branch hazard.
#198	**Q198: In the unoptimized five-stage pipeline described, during which stage is the decision about whether to branch made?**
	Answer: The branch decision occurs during the MEM (memory access) pipeline stage.
#199	**Q199: The delay in determining the proper instruction to fetch due to a branch instruction is called a ______ or branch hazard.**
	Answer: control hazard
#200	**Q200: Why is the section on control hazards presented as shorter than the one on data hazards?**
	Answer: Control hazards are simpler to understand, occur less frequently, and have less effective resolution schemes compared to forwarding for data hazards.
#201	**Q201: What is a simple static branch prediction scheme that improves upon stalling?**
	Answer: The 'assume branch not taken' scheme, where execution continues down the sequential instruction stream.
#202	**Q202: What action is taken to discard instructions in a pipeline when a branch is mispredicted?**
	Answer: The original control values for the incorrect instructions are changed to 0s.

#203	**Q203: Term: Flush**
	Answer: Definition: To discard instructions in a pipeline, usually due to an unexpected event like a mispredicted branch.
#204	**Q204: In the unoptimized pipeline, which stages must be flushed when a branch taken in the MEM stage is discovered?**
	Answer: The instructions in the IF, ID, and EX stages must be flushed.
#205	**Q205: What is one way to improve conditional branch performance by reducing the cost of a taken branch?**
	Answer: Move the conditional branch execution earlier in the pipeline, such as to the ID stage.
#206	**Q206: What are the two actions that must occur earlier to move branch execution from the MEM stage to the ID stage?**
	Answer: Computing the branch target address and evaluating the branch decision.
#207	**Q207: How is the branch target address calculation moved to the ID stage?**
	Answer: The branch adder is moved from the EX stage to the ID stage, using the PC and immediate field from the IF/ID register.
#208	**Q208: What is the more difficult part of moving branch execution to the ID stage?**
	Answer: Moving the branch decision itself, which involves comparing two register values during the ID stage.
#209	**Q209: Moving the branch test to the ID stage introduces the need for new _____ and _____ hardware.**
	Answer: forwarding, hazard detection
#210	**Q210: If branch comparison is moved to ID, from which pipeline registers can the bypassed source operands come?**
	Answer: The bypassed operands can come from either the EX/MEM or MEM/WB pipeline registers.
#211	**Q211: If branch execution is moved to the ID stage, what happens if an ALU instruction immediately preceding a branch produces an operand for the branch test?**
	Answer: A data hazard occurs, and a one-cycle stall is required.
#212	**Q212: If branch execution is moved to the ID stage, what is the penalty for a load instruction immediately followed by a branch that depends on the load result?**
	Answer: A two-cycle stall is needed.

#213	**Q213: What is the primary benefit of moving conditional branch execution to the ID stage?**
	Answer: It reduces the penalty of a taken branch to only one instruction.
#214	**Q214: What is the purpose of the IF.Flush control line?**
	Answer: It zeros the instruction field of the IF/ID pipeline register to flush the instruction currently in the IF stage.
#215	**Q215: Clearing the IF/ID register with IF.Flush transforms the fetched instruction into a _____, an instruction that has no action.**
	Answer: nop
#216	**Q216: What is the term for predicting branches at runtime using runtime information?**
	Answer: Dynamic branch prediction.
#217	**Q217: Is a prediction from a simple branch prediction buffer guaranteed to be for the correct branch instruction?**
	Answer: No, another branch with the same low-order address bits may have set the prediction bit, but this does not affect correctness.
#218	**Q218: What happens in a 1-bit dynamic prediction scheme when a prediction turns out to be wrong?**
	Answer: The incorrectly predicted instructions are deleted, the prediction bit is inverted, and the proper sequence is fetched.
#219	**Q219: What is the performance shortcoming of a simple 1-bit prediction scheme, especially with loops?**
	Answer: It can mispredict twice for a frequently-taken loop: once on the first iteration and once on the final, non-taken iteration.
#220	**Q220: For a loop branch that is taken 9 times and not taken once, what is the prediction accuracy of a 1-bit scheme?**
	Answer: The accuracy is 80%, with two incorrect predictions (first and last iteration) and eight correct ones.
#221	**Q221: How does a 2-bit branch prediction scheme improve upon a 1-bit scheme?**
	Answer: A prediction must be wrong twice before it is changed, making it more resilient to single changes in branch behavior.

#222	**Q222: In a 2-bit prediction scheme, a branch that strongly favors taken or not taken will be mispredicted how many times when its pattern is briefly interrupted?**
	Answer: It will be mispredicted only once.

#223	**Q223: A 2-bit scheme is a general instance of a _____ predictor, which is incremented on an accurate prediction and decremented otherwise.**
	Answer: counter-based

#224	**Q224: Term: Branch Target Buffer**
	Answer: Definition: A structure that caches the destination PC or destination instruction for a branch to reduce or eliminate the taken-branch penalty.

#225	**Q225: What kind of predictor combines the local behavior of a branch with global information about recently executed branches?**
	Answer: A correlating predictor.

#226	**Q226: Term: Correlating Predictor**
	Answer: Definition: A branch predictor that combines local behavior of a particular branch and global information about the behavior of some recent number of executed branches.

#227	**Q227: Term: Tournament Branch Predictor**
	Answer: Definition: A predictor with multiple prediction mechanisms for each branch and a selector that chooses which one to use based on which has been more accurate.

#228	**Q228: What is an architectural alternative to conditional branches that can reduce their frequency?**
	Answer: Conditional move instructions, which conditionally change a destination register's value instead of the PC.

#229	**Q229: What is the function of the ARMv8 CSEL instruction?**
	Answer: It conditionally selects one of two source registers to copy to a destination register based on the condition codes.

#230	**Q230: Given a 2-cycle misprediction penalty and 0-cycle correct prediction penalty, which scheme is best for a branch taken with 5% frequency?**
	Answer: Predict not taken is best, as it is correct 95% of the time, yielding a lower average penalty than the 90% accurate dynamic predictor.

#231	**Q231: Given a 2-cycle misprediction penalty and 0-cycle correct prediction penalty, which scheme is best for a branch taken with 95% frequency?**
	Answer: Predict taken is best, as it is correct 95% of the time, yielding a lower average penalty than the 90% accurate dynamic predictor.

#232	**Q232: Given a 2-cycle misprediction penalty and 0-cycle correct prediction penalty, which scheme is best for a branch taken with 70% frequency?**
	Answer: Dynamic prediction is best, as its 90% accuracy is better than both predict taken (70%) and predict not taken (30%).

Chapter 5.1 - Memory Hierarchy & Locality Principle

#233	**Q233: What illusion do memory hierarchies create for programmers?**
	Answer: They create the illusion of having unlimited amounts of fast memory.

#234	**Q234: What fundamental principle underlies the operation of memory hierarchies?**
	Answer: The principle of locality.

#235	**Q235: What does the principle of locality state about program behavior?**
	Answer: It states that programs access a relatively small portion of their address space at any instant of time.

#236	**Q236: What are the two different types of locality?**
	Answer: Temporal locality (locality in time) and spatial locality (locality in space).

#237	**Q237: Term: Temporal locality**
	Answer: The principle stating that if a data location is referenced, it will tend to be referenced again soon.

#238	**Q238: What common program structure demonstrates high temporal locality?**
	Answer: Loops, where instructions and data are accessed repeatedly.

#239	**Q239: Term: Spatial locality**
	Answer: The principle stating that if a data location is referenced, data locations with nearby addresses will tend to be referenced soon.

#240	**Q240: How does sequential access of instructions in a program demonstrate spatial locality?**
	Answer: Instructions are normally accessed sequentially, meaning that referencing one instruction makes it highly likely the next one in memory will be referenced soon.

#241	**Q241: Sequential access to elements of an array is an example of high _____ locality.**
	Answer: spatial

#242	**Q242: How do computer systems take advantage of the principle of locality?**
	Answer: By implementing the memory of a computer as a memory hierarchy.

#243	**Q243: Term: Memory hierarchy**
	Answer: A structure that uses multiple levels of memories where size and access time increase with distance from the processor.

#244	**Q244: What is the relationship between speed, size, and cost per bit in a memory hierarchy?**
	Answer: Faster memories are more expensive per bit and are therefore smaller.

#245	**Q245: What is the overall goal of implementing a memory hierarchy?**
	Answer: To provide memory as large as the cheapest level, with an access speed offered by the fastest level.

#246	**Q246: In a memory hierarchy, a level closer to the processor is generally a _____ of any level further away.**
	Answer: subset

#247	**Q247: In a memory hierarchy, data is copied between only _____ adjacent levels at a time.**
	Answer: two

#248	**Q248: Term: Block (or line)**
	Answer: The minimum unit of information that can be either present or not present in a level of the memory hierarchy.

#249	**Q249: What is it called when data requested by the processor is found in the upper level of the memory hierarchy?**
	Answer: A hit.

#250	**Q250: What is it called when data requested by the processor is not found in the upper level of the memory hierarchy?**
	Answer: A miss.

#251	**Q251: What action is taken in a memory hierarchy when a miss occurs?**
	Answer: The lower level in the hierarchy is accessed to retrieve the block containing the requested data.

#252	**Q252: Term: Hit rate**
	Answer: The fraction of memory accesses found in a level of the memory hierarchy.

#253	**Q253: Term: Miss rate**
	Answer: The fraction of memory accesses not found in a level of the memory hierarchy.

#254	**Q254: How is the miss rate related to the hit rate?**
	Answer: The miss rate is equal to 1 minus the hit rate.

#255	**Q255: Term: Hit time**
	Answer: The time required to access a level of the memory hierarchy, including the time to determine if the access is a hit or a miss.

#256	**Q256: Term: Miss penalty**
	Answer: The time required to replace a block in an upper level with the corresponding block from the lower level and deliver it.

#257	**Q257: Why is the hit time much smaller than the miss penalty?**
	Answer: Because the upper level is smaller and built with faster memory parts than the lower level.

#258	**Q258: Memory hierarchies keep more recently accessed data closer to the processor to take advantage of _____ locality.**
	Answer: temporal

#259	**Q259: How do memory hierarchies leverage spatial locality?**
	Answer: By moving blocks consisting of multiple contiguous words in memory to upper levels of the hierarchy.

#260	**Q260: In a true memory hierarchy, if data is present in level 'i', where else must it also be present?**
	Answer: It must also be present in the next lower level, 'i + 1'.

#261	**Q261: According to the 'Check Yourself' section, which of the following is generally true: 'Most of the cost of the memory hierarchy is at the highest level.'?**
	Answer: This statement is generally true.

#262	**Q262: According to the 'Check Yourself' section, which of the following is generally true: 'Most of the capacity of the memory hierarchy is at the lowest level.'?**
	Answer: This statement is generally true.

#263	**Q263: In the library analogy for memory hierarchies, what do the books on your desk represent?**
	Answer: The upper, faster level of the memory hierarchy (e.g., a cache).

#264	**Q264: In the library analogy, going back to the shelves to find a book corresponds to what event in a memory hierarchy?**
	Answer: A miss, which requires accessing the lower level of memory.

#265	**Q265: In the library analogy, what does finding the information you need in a book already on your desk represent?**
	Answer: A hit in the memory hierarchy.

#266	**Q266: What fundamental principle states that programs access a relatively small portion of their address space at any instant of time?**
	Answer: The principle of locality.

#267	**Q267: The locality principle stating that if an item is referenced, it will tend to be referenced again soon is known as _____ locality.**
	Answer: temporal

#268	**Q268: What common programming structure results in large amounts of temporal locality?**
	Answer: Loops, which cause instructions and data to be accessed repeatedly.

#269	**Q269: The locality principle stating that if an item is referenced, items whose addresses are close by will tend to be referenced soon is known as _____ locality.**
	Answer: spatial

#270	**Q270: What is an example of program behavior that exhibits high spatial locality?**
	Answer: Sequential access of instructions or sequential access to elements of an array.

#271	**Q271: How do memory hierarchies take advantage of temporal locality?**
	Answer: By keeping more recently accessed data items closer to the processor.

#272	**Q272: How do memory hierarchies take advantage of spatial locality?**
	Answer: By moving blocks of multiple contiguous words to upper levels of the hierarchy.

#273	**Q273: In a memory hierarchy, what happens to the size of the memories and the access time as the distance from the processor increases?**
	Answer: Both the size of the memories and the access time increase.

#274	**Q274: What is the primary goal of using a memory hierarchy?**
	Answer: To provide memory as large as the cheapest technology, with access speeds offered by the fastest memory.

#275	**Q275: What is the relationship between data in a higher level of the memory hierarchy and data in a lower level?**
	Answer: A level closer to the processor is generally a subset of any level further away.

#276	**Q276: In most systems, if data is present in memory level 'i', where else must it be present?**
	Answer: It must also be present in level 'i + 1'.

#277	**Q277: In a memory hierarchy, data is copied between only two _____ levels at a time.**
	Answer: adjacent

#278	**Q278: What is a `hit` in the context of a memory hierarchy?**
	Answer: A memory access where the requested data is found in the upper level of the hierarchy.

#279	**Q279: What is a `miss` in the context of a memory hierarchy?**
	Answer: A memory access where the requested data is not found in the upper level of the hierarchy.

#280	**Q280: What action is taken in a memory hierarchy when a `miss` occurs?**
	Answer: The lower level in the hierarchy is accessed to retrieve the block containing the requested data.

#281	**Q281: How is the `miss rate` related to the `hit rate`?**
	Answer: The miss rate is equal to 1 minus the hit rate (1 - hit rate).

#282	**Q282: What is the major component of the miss penalty?**
	Answer: The time to access the next, lower level in the hierarchy.

#283	**Q283: Why must modern programmers understand the memory hierarchy?**
	Answer: To write code that achieves good performance, as memory access is a major factor.

#284	**Q284: With a high enough hit rate, a memory hierarchy has an effective access time close to that of the _____ level.**
	Answer: highest (and fastest)

#285	**Q285: A memory hierarchy gives the user the illusion of a memory size equal to that of the _____ level.**
	Answer: lowest (and largest)

#286	**Q286: In the library analogy, what do the books on the desk represent?**
	Answer: The upper, faster level of the memory hierarchy (e.g., the cache).

#287	**Q287: In the library analogy, what does going back to the shelves to find a book represent?**
	Answer: A 'miss' that requires accessing the lower, slower level of the memory hierarchy.

#288	**Q288: In the library analogy, what does a single book represent?**
	Answer: A 'block' or 'line' of information.

#289	**Q289: According to the 'Check Yourself' section, which statement is generally true: 'Most of the cost of the memory hierarchy is at the highest level'?**
	Answer: False, the highest level is smallest, so most cost is in the larger, lower levels.

#290	**Q290: According to the 'Check Yourself' section, which statement is generally true: 'Most of the capacity of the memory hierarchy is at the lowest level'?**
	Answer: True, because the lowest levels are the largest.

Chapter 5.2 - Memory Technologies (SRAM, DRAM, Flash & Disk)

#291	**Q291: What type of memory is typically used to implement main memory in modern computer hierarchies?**
	Answer: DRAM (dynamic random access memory).

#292	**Q292: What type of memory is typically used for levels closer to the processor, such as caches?**
	Answer: SRAM (static random access memory).

#293	**Q293: Which is less costly per bit, DRAM or SRAM?**
	Answer: DRAM is less costly per bit than SRAM.

#294	**Q294: What is the primary reason for the price difference between DRAM and SRAM?**
	Answer: DRAM uses significantly less area per bit, allowing for larger capacity on the same amount of silicon.

#295	**Q295: What nonvolatile memory technology is commonly used as secondary memory in Personal Mobile Devices?**
	Answer: Flash memory.

#296	**Q296: In servers, what technology is used to implement the largest and slowest level in the memory hierarchy?**
	Answer: Magnetic disk.

#297	**Q297: According to the 2012 data, what is the typical access time for SRAM?**
	Answer: 0.5–2.5 nanoseconds.

#298	**Q298: According to the 2012 data, what is the typical access time for DRAM?**
	Answer: 50–70 nanoseconds.

#299	**Q299: According to the 2012 data, what is the typical access time for Flash memory?**
	Answer: 5,000–50,000 nanoseconds.
#300	**Q300: According to the 2012 data, what is the typical access time for a magnetic disk?**
	Answer: 5,000,000–20,000,000 nanoseconds.
#301	**Q301: Why is the access time for SRAM very close to its cycle time?**
	Answer: Because SRAMs do not need to be refreshed.
#302	**Q302: How many transistors per bit do SRAMs typically use?**
	Answer: Six to eight transistors per bit.
#303	**Q303: What is the primary function of the six to eight transistors in an SRAM cell?**
	Answer: To prevent the stored information from being disturbed when read.
#304	**Q304: Due to Moore's Law, where are all levels of caches typically integrated today?**
	Answer: They are integrated onto the processor chip.
#305	**Q305: How is a value stored in a DRAM cell?**
	Answer: As a charge in a capacitor.
#306	**Q306: How many transistors are used per bit of storage in a DRAM cell?**
	Answer: A single transistor.
#307	**Q307: Why is DRAM so much denser and cheaper per bit than SRAM?**
	Answer: Because DRAMs use only one transistor per bit of storage, compared to SRAM's six to eight.
#308	**Q308: Why is DRAM called 'dynamic'?**
	Answer: Because the charge stored in its capacitor cells cannot be kept indefinitely and must be periodically refreshed.

#309	**Q309: What is the process for refreshing a DRAM cell?**
	Answer: Its contents are read and then written back.

#310	**Q310: How does the two-level decoding structure in DRAMs facilitate efficient refreshing?**
	Answer: It allows an entire row to be refreshed at once with a single read cycle followed by a write cycle.

#311	**Q311: In a modern DRAM's internal organization, what command is sent to open or close a bank?**
	Answer: A PRE (precharge) command.

#312	**Q312: In a modern DRAM, what command causes a row to be transferred to a buffer?**
	Answer: An Act (activate) command, which is sent with the row address.

#313	**Q313: How does a DRAM improve performance for repeated accesses to the same row?**
	Answer: It buffers the entire row, and this buffer acts like an SRAM, allowing for much faster access to bits within that row.

#314	**Q314: What is the primary advantage of Synchronous DRAMs (SDRAMs)?**
	Answer: The use of a clock eliminates the time needed for the memory and processor to synchronize.

#315	**Q315: How do SDRAMs achieve a speed advantage for burst transfers?**
	Answer: The clock transfers successive bits in a burst without needing additional address bits to be specified.

#316	**Q316: What does DDR stand for in DDR SDRAM?**
	Answer: Double Data Rate.

#317	**Q317: What is the key mechanism behind Double Data Rate (DDR) technology?**
	Answer: Data is transferred on both the rising and falling edges of the clock, effectively doubling the bandwidth.

#318	**Q318: A DDR4-3200 DRAM can perform 3200 million transfers per second. What is its clock frequency?**
	Answer: 1600 MHz.

#319	**Q319: What internal DRAM organization technique allows for simultaneous reads or writes from different sections to increase bandwidth?**
	Answer: Organizing the DRAM into multiple banks, each with its own row buffer.

#320	**Q320: Term: Address Interleaving**
	Answer: A rotating access scheme where an address is sent to several banks, allowing them all to read or write simultaneously to supply higher bandwidth.

#321	**Q321: What is a DIMM?**
	Answer: A dual inline memory module, which is a small board containing multiple DRAM chips, commonly used for server memory.

#322	**Q322: A DIMM using DDR4-3200 SDRAMs, organized to be 8 bytes wide, would be named what, based on its bandwidth?**
	Answer: PC25600.

#323	**Q323: Term: Memory Rank**
	Answer: A subset of chips in a DIMM that share common address lines.

#324	**Q324: Flash memory is a type of _____.**
	Answer: electrically erasable programmable read-only memory (EEPROM)

#325	**Q325: What is a primary limitation of flash memory regarding write operations?**
	Answer: Writes can wear out the flash memory bits over time.

#326	**Q326: Term: Wear Leveling**
	Answer: A technique used by flash memory controllers to spread writes evenly by remapping frequently written blocks to less used blocks.

#327	**Q327: What is the main purpose of wear leveling in flash memory products?**
	Answer: To prevent premature failure of the memory by distributing the wear from write operations across all blocks.

#328	**Q328: What are the rotating components of a magnetic hard disk called?**
	Answer: Platters.

#329	**Q329: What is the component in a hard disk that contains a small electromagnetic coil to read and write information?**
	Answer: A read-write head.

#330	**Q330: Term: Track (on a magnetic disk)**
	Answer: One of thousands of concentric circles that make up the surface of a magnetic disk.

#331	**Q331: Term: Sector (on a magnetic disk)**
	Answer: One of the segments that make up a track; it is the smallest amount of information that is read or written on a disk.

#332	**Q332: Term: Cylinder (on a magnetic disk)**
	Answer: The set of all tracks under the read-write heads at a given point on all surfaces.

#333	**Q333: What is the first stage in a disk access process, involving positioning the head over the proper track?**
	Answer: A seek.

#334	**Q334: Term: Seek Time**
	Answer: The time required to move the read-write head to the desired track on a disk.

#335	**Q335: Why might the actual average seek time be much lower than the advertised average seek time?**
	Answer: Due to the locality of disk references, where successive accesses are often to the same file or nearby tracks.

#336	**Q336: Term: Rotational Latency**
	Answer: The time required for the desired sector of a disk to rotate under the read-write head.

#337	**Q337: What is the average rotational latency generally assumed to be?**
	Answer: Half the time of a full rotation.

#338	**Q338: What is the third and final component of a disk access time?**
	Answer: Transfer time, which is the time to transfer the actual block of bits.

#339	**Q339: Transfer time is a function of what three factors?**
	Answer: Sector size, rotation speed, and the recording density of a track.

#340	**Q340: How do modern high-level disk interfaces organize logical blocks to speed up sequential transfers?**
	Answer: They are ordered in a serpentine fashion across a surface, rather than strictly following the old sector-track-cylinder model.

#341	**Q341: Compared to DRAM, approximately how much faster is Flash memory?**
	Answer: Flash memory is not faster than DRAM; DRAM is approximately 100,000 times faster than a magnetic disk, while Flash is 1000 times faster.

#342	**Q342: What is a key advantage of magnetic disks over flash memory regarding data writes?**
	Answer: Magnetic disks do not have a write wear-out problem.

#343	**Q343: Why is flash memory a better match than magnetic disks for personal mobile devices?**
	Answer: Flash memory is much more rugged and can better withstand the jostling inherent in mobile devices.

#344	**Q344: In a memory hierarchy, the _____ acts as a safe place to store things that are needed frequently.**
	Answer: cache

#345	**Q345: Term: Cache**
	Answer: Any storage managed to take advantage of locality of access, most commonly the level of memory between the processor and main memory.

#346	**Q346: What is a 'cache miss'?**
	Answer: An event that occurs when a requested data item is not found in the cache, requiring it to be fetched from a lower level of memory.

#347	**Q347: Term: Direct-mapped cache**
	Answer: A cache structure in which each memory location is mapped to exactly one location in the cache.

#348	**Q348: How is the specific cache location determined for a memory address in a typical direct-mapped cache?**
	Answer: By using the formula: (Block address) modulo (Number of blocks in the cache).
#349	**Q349: In a direct-mapped cache with a size that is a power of 2, which bits of the address are used as the index?**
	Answer: The low-order bits of the address.
#350	**Q350: Term: Tag (in a cache)**
	Answer: A field in the cache that contains the address information required to identify whether the data in a cache block corresponds to a requested word.
#351	**Q351: What part of the memory address is stored in the cache tag?**
	Answer: The upper portion of the address, corresponding to the bits that are not used as an index into the cache.
#352	**Q352: What is the purpose of a 'valid bit' in a cache?**
	Answer: To indicate whether a cache block contains valid information or not (e.g., after a processor starts up).
#353	**Q353: What are the four primary technologies used in modern memory hierarchies?**
	Answer: DRAM (Dynamic Random Access Memory), SRAM (Static Random Access Memory), Flash memory, and Magnetic Disk.
#354	**Q354: In a typical memory hierarchy, what technology is used for main memory?**
	Answer: DRAM (Dynamic Random Access Memory).
#355	**Q355: In a typical memory hierarchy, what technology is used for cache levels closer to the processor?**
	Answer: SRAM (Static Random Access Memory).
#356	**Q356: Which is more costly per bit, SRAM or DRAM?**
	Answer: SRAM is significantly more costly per bit than DRAM.
#357	**Q357: What is the primary reason for the price difference between SRAM and DRAM?**
	Answer: DRAM uses significantly less area per bit of memory, allowing for larger capacity on the same amount of silicon.

#358	**Q358: What type of memory is typically used as secondary memory in Personal Mobile Devices (PMDs)?**
	Answer: Flash memory.

#359	**Q359: In servers, what technology is used to implement the largest and slowest level of the memory hierarchy?**
	Answer: Magnetic disk.

#360	**Q360: What was the typical access time for SRAM in 2012?**
	Answer: 0.5 to 2.5 nanoseconds.

#361	**Q361: What was the typical access time for DRAM in 2012?**
	Answer: 50 to 70 nanoseconds.

#362	**Q362: What was the typical access time for Flash memory in 2012?**
	Answer: 5,000 to 50,000 nanoseconds.

#363	**Q363: What was the typical access time for a magnetic disk in 2012?**
	Answer: 5,000,000 to 20,000,000 nanoseconds.

#364	**Q364: _____s have a fixed access time to any datum, though the read and write access times may differ.**
	Answer: SRAM

#365	**Q365: How many transistors do SRAMs typically use per bit of storage?**
	Answer: Six to eight transistors per bit.

#366	**Q366: Why do SRAMs not need to be refreshed?**
	Answer: As long as power is applied, the value can be kept indefinitely without the charge leaking away.

#367	**Q367: What is the primary modern application for SRAM technology?**
	Answer: All levels of caches integrated directly onto the processor chip.

#368	**Q368: How is a value stored in a Dynamic RAM (DRAM) cell?**
	Answer: As a charge in a capacitor.

#369	**Q369: The need for a DRAM cell to be periodically _____ is why this memory structure is called dynamic.**
	Answer: refreshed

#370	**Q370: How does DRAM's two-level decoding structure make refreshing efficient?**
	Answer: It allows an entire row to be refreshed at once with a single read and write cycle.

#371	**Q371: In a modern DRAM, what does sending an Act (activate) command with a row address accomplish?**
	Answer: It causes the entire specified row to be transferred to a buffer.

#372	**Q372: How do DRAMs improve performance for repeated accesses to the same row?**
	Answer: They buffer the entire row, and this buffer acts like a faster SRAM for subsequent accesses within that row.

#373	**Q373: What is the main advantage of Synchronous DRAMs (SDRAMs)?**
	Answer: The use of a clock eliminates the time needed for the memory and processor to synchronize.

#374	**Q374: What does the name 'Double Data Rate (DDR) SDRAM' signify?**
	Answer: Data is transferred on both the rising and falling edges of the clock, doubling the bandwidth for a given clock rate.

#375	**Q375: A DDR4-3200 DRAM can perform 3200 million transfers per second. What is its clock rate?**
	Answer: 1600 MHz.

#376	**Q376: What is the name of the technique that uses multiple internal DRAM banks to increase bandwidth by rotating accesses between them?**
	Answer: Address interleaving.

#377	**Q377: Term: DIMM**
	Answer: Definition: Dual Inline Memory Module, a small board containing multiple DRAM chips, commonly used for server memory.

#378	**Q378: How is the name of a DIMM, such as PC25600, derived?**
	Answer: It is named after its bandwidth in megabytes per second (e.g., a DDR4-3200 DIMM that is 8 bytes wide has a bandwidth of 25,600 MB/s).

#379	**Q379: What term is used for the subset of chips on a DIMM that share common address lines and are used for a particular transfer?**
	Answer: Memory rank.

#380	**Q380: Flash memory is a type of _____ memory.**
	Answer: electrically erasable programmable read-only memory (EEPROM)

#381	**Q381: What is the primary durability issue with flash memory?**
	Answer: Writes can wear out the memory bits over time.

#382	**Q382: What is the name of the technique used in flash memory to spread writes across different blocks to prevent premature wear?**
	Answer: Wear leveling.

#383	**Q383: How does wear leveling work in flash memory controllers?**
	Answer: It remaps blocks that have been written many times to less frequently used blocks.

#384	**Q384: A magnetic hard disk consists of a collection of _____, which rotate on a spindle.**
	Answer: platters

#385	**Q385: What is the name of the small electromagnetic coil on a movable arm used to read and write information on a hard disk?**
	Answer: A read-write head.

#386	**Q386: What is a 'cylinder' in the context of a magnetic disk?**
	Answer: The set of all tracks that are under the read-write heads at a given point on all surfaces.

#387	**Q387: What are the three main stages of a disk access process?**
	Answer: Seek time, rotational latency, and transfer time.

#388	**Q388: Term: Seek (in disk access)**
	Answer: Definition: The process of positioning a read/write head over the proper track on a disk.
#389	**Q389: The time required to move the disk head to the desired track is called the ____ time.**
	Answer: seek
#390	**Q390: On average, how much of a full circle must a disk platter spin to reach the desired sector?**
	Answer: Half a rotation.
#391	**Q391: What is the final component of a disk access, which is a function of sector size, rotation speed, and recording density?**
	Answer: Transfer time.
#392	**Q392: What is a primary difference between magnetic disks and semiconductor memory like DRAM in terms of access time?**
	Answer: Disks have a much slower access time because they are mechanical devices (DRAM is ~100,000 times faster).
#393	**Q393: What is a primary difference between magnetic disks and semiconductor memory like DRAM in terms of cost?**
	Answer: Disks are significantly cheaper per bit (10 to 100 times cheaper).
#394	**Q394: Unlike flash memory, magnetic disks do not have a _____ problem.**
	Answer: write wear-out
#395	**Q395: Why is flash memory a better match for personal mobile devices than magnetic disks?**
	Answer: Flash is much more rugged and can better withstand the jostling inherent in mobile devices.
#396	**Q396: What is a 'direct-mapped cache'?**
	Answer: A cache structure in which each memory location is mapped to exactly one location in the cache.
#397	**Q397: How is the cache location for a memory address typically determined in a direct-mapped cache?**
	Answer: By using the formula: (Block address) modulo (Number of blocks in the cache).

#398	**Q398: In a cache, what is the purpose of a 'tag'?**
	Answer: It contains the address information required to identify whether the data in a cache block corresponds to the requested word.

Chapter 5.3 - Cache Basics & Direct-Mapped Caches

#399	**Q399: What is the general term for any storage managed to take advantage of locality of access?**
	Answer: A cache.

#400	**Q400: What is the event called when a processor requests a word that is not in the cache, and what is the immediate result?**
	Answer: It is called a cache miss, and it forces the cache to fetch the requested word from memory.

#401	**Q401: What is a `direct-mapped cache`?**
	Answer: A cache structure in which each memory location is mapped to exactly one location in the cache.

#402	**Q402: What formula is typically used to map a memory block address to a location in a direct-mapped cache?**
	Answer: (Block address) modulo (Number of blocks in the cache).

#403	**Q403: How is the modulo operation for address mapping simplified if the number of entries in the cache is a power of 2?**
	Answer: By using the low-order $log_2(\text{cache size in blocks})$ bits of the address as the index.

#404	**Q404: What is the purpose of the `tag` field in a cache entry?**
	Answer: It contains the address information required to identify whether a block in the cache corresponds to a requested word.

#405	**Q405: Which portion of the memory address is stored in the cache entry's tag field?**
	Answer: The upper portion of the address, corresponding to the bits that are not used as the cache index.

#406	**Q406: What is the purpose of the `valid bit` in a cache entry?**
	Answer: It indicates whether the associated block in the cache contains valid data.

#407	**Q407: When a processor first starts up, what is the typical state of the cache's valid bits?**
	Answer: They are not set (e.g., 'N' or 0), indicating that the cache entries do not contain valid information.
#408	**Q408: Caching is perhaps the most important example of what major computer architecture concept?**
	Answer: Prediction.
#409	**Q409: Caching relies on the principle of _____ to attempt to find desired data in higher levels of the memory hierarchy.**
	Answer: locality
#410	**Q410: A referenced memory address is divided into which two primary fields for a direct-mapped cache lookup?**
	Answer: A tag field and a cache index.
#411	**Q411: What two components of a cache entry uniquely specify the memory address of the word contained in that cache block?**
	Answer: The cache index and the tag contents of that block.
#412	**Q412: Why must the total number of entries in a direct-mapped cache be a power of 2?**
	Answer: Because the index field is used as an address to reference the cache, and an n-bit field has 2^n possible values.
#413	**Q413: For a cache with a one-word block size, how is a memory address with 64 bits, 2^n blocks, and 2^m words per block broken down?**
	Answer: The tag field is $64 - n - m - 2$ bits, the index is n bits, the block offset is m bits, and the byte offset is 2 bits.
#414	**Q414: By convention, when a cache is referred to by its size (e.g., a "4 KiB cache"), what components are excluded from this measurement?**
	Answer: The storage for the tag and valid fields; only the data storage is counted.
#415	**Q415: For a cache with 64 blocks and a block size of 16 bytes, to what block number does byte address 1200 map?**
	Answer: Block number 11, which is calculated as $(\lfloor 1200 / 16 \rfloor) \pmod{64} = 75 \pmod{64} = 11$.

#416	**Q416: How do larger block sizes in a cache exploit spatial locality?**
	Answer: By fetching adjacent words along with the requested word, they decrease the miss rate for subsequent sequential accesses.

#417	**Q417: What is a potential negative consequence of making a cache's block size too large relative to the total cache size?**
	Answer: The miss rate may increase due to a high degree of competition for the few available blocks.

#418	**Q418: What is the primary drawback of increasing cache block size in terms of performance?**
	Answer: It increases the miss penalty because more time is required to transfer the larger block from memory.

#419	**Q419: What are the two components that determine the time required to fetch a block during a cache miss?**
	Answer: The latency to the first word and the transfer time for the rest of the block.

#420	**Q420: Which general guideline is valid for a cache designer concerning memory bandwidth?**
	Answer: The higher the memory bandwidth, the larger the cache block size can be.

#421	**Q421: What is a `cache miss`?**
	Answer: A request for data from the cache that cannot be filled because the data is not present in the cache.

#422	**Q422: The processing of a cache miss creates a _____ in the processor, freezing the contents of registers while waiting for memory.**
	Answer: pipeline stall

#423	**Q423: What is the first step in handling an instruction cache miss?**
	Answer: Send the original Program Counter (PC) value to the memory system.

#424	**Q424: After sending the address, what is the second step in handling an instruction cache miss?**
	Answer: Instruct main memory to perform a read and wait for the access to complete.

#425	**Q425: What is the third step in handling an instruction cache miss, after the data is returned from memory?**
	Answer: Write the data into the cache, write the upper address bits into the tag field, and set the valid bit to on.

#426	**Q426: What is the final step in handling an instruction cache miss?**
	Answer: Restart the instruction execution, which will refetch the instruction, causing a cache hit.

#427	**Q427: If a write operation modifies only the data cache without changing main memory, the cache and memory are said to be _____.**
	Answer: inconsistent

#428	**Q428: What is the `write-through` scheme for handling writes?**
	Answer: A scheme in which writes always update both the cache and the next lower level of the memory hierarchy.

#429	**Q429: What is the primary performance drawback of a simple write-through scheme?**
	Answer: Every write incurs the long latency of an access to main memory, which can significantly slow down the processor.

#430	**Q430: What is a `write buffer`?**
	Answer: A queue that holds data while it is waiting to be written to memory, allowing the processor to continue execution.

#431	**Q431: When using a write buffer, under what condition must the processor stall during a write operation?**
	Answer: The processor must stall if the write buffer is full.

#432	**Q432: What is the `write-back` scheme for handling writes?**
	Answer: A scheme where a write updates only the block in the cache, and the modified block is written to memory only when it is replaced.

#433	**Q433: What is the `write allocate` policy for a write miss?**
	Answer: The block is fetched from memory and placed into the cache, and then the relevant word is overwritten.

#434	**Q434: What is the `no write allocate` policy for a write miss?**
	Answer: The word is updated in main memory, but the block is not brought into the cache.

#435	**Q435: What is the primary motivation for using a `no write allocate` policy?**
	Answer: To avoid unnecessary memory fetches when a program writes an entire block of data, such as when an OS zeros a page.

#436	**Q436: Why is implementing stores more complex in a write-back cache compared to a write-through cache?**
	Answer: The cache block cannot be overwritten until a hit is confirmed, because memory does not have a correct copy of the potentially modified data.

#437	**Q437: What is a `split cache`?**
	Answer: A memory hierarchy level composed of two independent caches, one for instructions and one for data, that operate in parallel.

#438	**Q438: What is the main advantage of using a split instruction and data cache?**
	Answer: It doubles the cache bandwidth by allowing simultaneous instruction and data accesses.

#439	**Q439: What is a potential disadvantage of a split cache compared to a single combined cache of the same total size?**
	Answer: The effective miss rate can be slightly higher due to the rigid partitioning of cache entries.

#440	**Q440: What is the `early restart` technique used for?**
	Answer: To reduce miss penalty by resuming execution as soon as the requested word of a block arrives, without waiting for the entire block.

#441	**Q441: What is the `critical word first` technique?**
	Answer: A scheme to reduce miss penalty where the memory system transfers the requested word first, then sends the rest of the block.

#442	**Q442: When a cache miss requires replacing an existing block in a direct-mapped cache, what principle of locality is being exploited?**
	Answer: Temporal locality, as a recently referenced item replaces a less recently referenced item.

#443	**Q443: In the provided example, why does a reference to address 18 (10010two) replace the entry for address 26 (11010two)?**
	Answer: Because both memory addresses map to the same cache index (010two) in the direct-mapped cache.

#444	**Q444: In the example of the 4 KiB cache in Figure 5.10, how many bits of the 64-bit address are used for the tag?**
	Answer: 52 bits, calculated as 64 total bits - 10 index bits - 2 byte-offset bits.

#445	**Q445: What two conditions must be met for a cache access to result in a hit?**
	Answer: The tag in the cache entry must match the tag portion of the address, and the entry's valid bit must be on.

#446	**Q446: How can a write buffer also be used with a write-back cache to reduce miss penalty?**
	Answer: When a miss replaces a modified block, the modified block can be moved to the buffer to be written to memory later, while the requested block is read.

#447	**Q447: How is the full memory address 10010two reconstructed from the cache entry at index 010two in Figure 5.9f?**
	Answer: By concatenating the tag field (10two) with the index field (010two).

#448	**Q448: In the context of computer memory hierarchy, what is a cache?**
	Answer: A small, fast storage level that acts as a safe place to store things (data) needed by the processor, positioned between the processor and main memory.

#449	**Q449: The term 'cache' is used to refer to any storage managed to take advantage of _____.**
	Answer: locality of access

#450	**Q450: What is the term for a processor's request for a word that is not currently in the cache?**
	Answer: A cache miss.

#451	**Q451: Following a cache miss, what action is taken with the requested data word?**
	Answer: The word is brought from the lower-level memory into the cache.

#452	**Q452: What is the simplest way to assign a location in the cache for each word in memory?**
	Answer: To assign the cache location based on the address of the word in memory.

#453	**Q453: Term: Direct-mapped cache**
	Answer: A cache structure in which each memory location is mapped to exactly one location in the cache.

#454	**Q454: What is the typical formula used by direct-mapped caches to determine the block location for a given memory address?**
	Answer: (Block address) modulo (Number of blocks in the cache).

#455	**Q455: In a direct-mapped cache where the number of entries is a power of 2, how is the mapping index computed from the address?**
	Answer: By using the low-order $log_2(ext{cache size in blocks})$ bits of the block address.

#456	**Q456: An 8-block direct-mapped cache uses the _____ lowest bits of the block address as the cache index.**
	Answer: 3

#457	**Q457: How does a cache distinguish between the multiple different memory locations that could map to the same cache entry?**
	Answer: By using tags, which store the upper portion of the address.

#458	**Q458: Term: Tag**
	Answer: A field in a memory hierarchy table that contains the address information required to identify if a block corresponds to a requested word.

#459	**Q459: What portion of the memory address must the tag field in a cache contain?**
	Answer: The upper portion of the address, corresponding to the bits not used as an index into the cache.

#460	**Q460: Why are the index bits of a memory address typically omitted from the cache tag?**
	Answer: They are redundant, as the index field by definition must match the cache block number being accessed.

#461	**Q461: What mechanism indicates that a cache block does not contain valid information, such as on processor start-up?**
	Answer: A valid bit.

#462	**Q462: Term: Valid bit**
	Answer: A field in the tables of a memory hierarchy that indicates that the associated block in the hierarchy contains valid data.

#463	**Q463: If a cache block's valid bit is not set, what is the outcome of a tag comparison for that block?**
	Answer: A match is impossible, resulting in a miss.

#464	**Q464: Caching is considered the most important example of what "big idea" in computer architecture?**
	Answer: Prediction.

#465	**Q465: In the example sequence of memory references, what event occurs on the eighth reference to address 18 (10010_{two})?**
	Answer: It causes a miss and must replace the word at address 26 (11010_{two}) in cache block 2 (010_{two}).

#466	**Q466: In a direct-mapped cache, what happens if a newly requested item maps to a block that is already occupied by valid data?**
	Answer: The existing data is replaced, as there is only one possible location for the new item.

#467	**Q467: For a simple cache lookup, a referenced memory address is primarily divided into which two fields?**
	Answer: A tag field and a cache index.

#468	**Q468: What is the function of the cache index derived from a memory address?**
	Answer: It is used to select a specific block (or entry) within the cache.

#469	**Q469: During a cache access, what is the tag field from the memory address used for?**
	Answer: It is compared with the tag value stored in the selected cache block to verify an address match.

#470	**Q470: What two conditions must be met for a cache access to result in a 'hit'?**
	Answer: The tag from the address must match the tag stored in the cache block, and the block's valid bit must be on.

#471	**Q471: The total number of entries in a direct-mapped cache must be a _____.**
	Answer: power of 2

#472	**Q472: For a 64-bit address system with a direct-mapped cache of 2^n blocks and a block size of 2^m words, what is the size of the tag field in bits?**
	Answer: The tag size is $64 - n - (m+2)$ bits.

#473	**Q473: What is the formula for the total number of bits in a direct-mapped cache with 2^n blocks?**
	Answer: $2^n \times (\text{block size in bits} + \text{tag size in bits} + 1 \text{ for valid bit})$.

#474	**Q474: When stating a cache's size (e.g., '16 KiB cache'), what parts are conventionally excluded from the total bit count?**
	Answer: The storage for tags and valid bits are excluded; only the data storage is counted.

#475	**Q475: For a cache with 16 KiB of data and four-word blocks, how many blocks does the cache contain?**
	Answer: 1024 blocks (16 KiB = 4096 words; 4096 words / 4 words/block = 1024 blocks).

#476	**Q476: To find the block address from a given byte address, what calculation is performed?**
	Answer: Block address = $\lfloor \frac{\text{Byte address}}{\text{Bytes per block}} \rfloor$.

#477	**Q477: Given a cache with 64 blocks and a 16-byte block size, to what block number does byte address 1200 map?**
	Answer: Block number 11 (since $(1200 / 16) \pmod{64} = 75 \pmod{64} = 11$).

#478	**Q478: What principle of locality do larger cache blocks primarily exploit to lower miss rates?**
	Answer: Spatial locality.

#479	**Q479: What is the typical effect on miss rate when increasing the cache block size?**
	Answer: It usually decreases the miss rate, up to a certain point.

#480	**Q480: What negative consequence can occur if the block size becomes a significant fraction of the total cache size?**
	Answer: The miss rate may increase due to heavy competition for a smaller number of available blocks.

#481	**Q481: What is the most significant drawback of increasing cache block size?**
	Answer: It increases the miss penalty, as fetching a larger block from memory takes more time.

#482	**Q482: Term: Cache miss**
	Answer: A request for data from the cache that cannot be filled because the data are not present in the cache.

#483	**Q483: A cache miss creates a _____, which freezes the processor while waiting for memory, as opposed to a full context-switching exception.**
	Answer: pipeline stall

#484	**Q484: What is the first step in the process of handling an instruction cache miss?**
	Answer: Send the original Program Counter (PC) value to the memory system.
#485	**Q485: What is the second step in handling an instruction cache miss?**
	Answer: Instruct main memory to perform a read and wait for the access to complete.
#486	**Q486: What is the third step in handling an instruction cache miss?**
	Answer: Write the new cache entry, loading data from memory, storing the upper address bits as the tag, and setting the valid bit.
#487	**Q487: What is the final step after a cache miss has been fully handled?**
	Answer: Restart the instruction execution at the first step, which will refetch the instruction (now from the cache).
#488	**Q488: If a store instruction updates only the cache and not main memory, the cache and memory are said to be _____.**
	Answer: inconsistent
#489	**Q489: Term: Write-through**
	Answer: A scheme in which writes always update both the cache and the next lower level of the memory hierarchy.
#490	**Q490: What is the major performance disadvantage of a simple write-through cache scheme?**
	Answer: Every write operation incurs the long latency of writing to main memory, which can significantly slow down the processor.
#491	**Q491: What hardware component is often used to mitigate the performance penalty of a write-through cache?**
	Answer: A write buffer.
#492	**Q492: Term: Write buffer**
	Answer: A queue that holds data while the data are waiting to be written to memory, allowing the processor to continue execution.
#493	**Q493: Under what condition must a processor stall even when using a write buffer?**
	Answer: The processor must stall if it needs to perform a write when the write buffer is already full.

#494	**Q494: Term: Write-back**
	Answer: A scheme that handles writes by updating values only to the block in the cache, then writing the modified block to memory only when it is replaced.

#495	**Q495: In a write-through cache, the strategy of fetching the block from memory on a write miss before overwriting the relevant portion is called _____.**
	Answer: write allocate

#496	**Q496: What is the 'no write allocate' policy for a write miss?**
	Answer: An alternative strategy where the portion of the block in memory is updated, but the block is not brought into the cache.

#497	**Q497: Why can't a write-back cache simply overwrite a block on a store instruction before confirming a hit?**
	Answer: Doing so could destroy the only valid copy of the data if the block was modified and has not yet been written back to memory.

#498	**Q498: What kind of cache organization does the Intrinsity FastMATH processor use to avoid stalling its pipeline?**
	Answer: Separate (split) instruction and data caches.

#499	**Q499: The Intrinsity FastMATH processor's data cache is 16 KiB, contains 256 blocks, and has a block size of _____ words.**
	Answer: 16

#500	**Q500: For the 32-bit Intrinsity FastMATH cache, what are the respective sizes of the tag, index, and block offset fields?**
	Answer: The tag is 18 bits, the index is 8 bits, and the block offset (word + byte) is 6 bits.

#501	**Q501: What two write policies does the Intrinsity FastMATH processor support?**
	Answer: It supports both write-through and write-back, selectable by the operating system.

#502	**Q502: Term: Split cache**
	Answer: A scheme where a memory hierarchy level has two independent caches operating in parallel, one for instructions and one for data.

#503	**Q503: What is the primary motivation for using split instruction and data caches in modern processors?**
	Answer: To increase cache bandwidth to meet the high demands of modern pipelines, allowing simultaneous instruction and data access.

#504	**Q504: How does the miss rate of a 32 KiB combined cache typically compare to the effective miss rate of two 16 KiB split caches?**
	Answer: The combined cache usually has a slightly lower (better) miss rate.

#505	**Q505: Why is the slightly higher miss rate of a split cache an acceptable trade-off?**
	Answer: The advantage of doubling the cache bandwidth far outweighs the small disadvantage of a slightly increased miss rate.

#506	**Q506: Using a cache block size larger than one word improves efficiency by reducing the amount of _____ storage relative to data storage.**
	Answer: tag

#507	**Q507: Which design guideline is generally valid regarding cache block size and memory bandwidth?**
	Answer: The higher the memory bandwidth, the larger the cache block can be.

#508	**Q508: What technique, called _____, reduces the effective miss penalty by resuming execution as soon as the specifically requested word of a block arrives from memory?**
	Answer: early restart

#509	**Q509: What memory transfer technique organizes the memory to send the requested word first, then the rest of the block, to reduce miss penalty?**
	Answer: Requested word first (or critical word first).

Chapter 5.4 - Cache Performance & Miss Penalty

#510	**Q510: What are the two primary techniques explored for improving cache performance?**
	Answer: Reducing the miss rate and reducing the miss penalty.

#511	**Q511: CPU time can be divided into the clock cycles the CPU spends executing the program and the clock cycles it spends _____.**
	Answer: waiting for the memory system

#512	**Q512: What is the formula for CPU time in terms of execution cycles and memory-stall cycles?**
	Answer: CPU time = (CPU execution clock cycles + Memory-stall clock cycles) × Clock cycle time.
#513	**Q513: Memory-stall clock cycles are primarily caused by what events?**
	Answer: Cache misses.
#514	**Q514: What is the formula for calculating Read-stall cycles?**
	Answer: Read-stall cycles = (Reads/Program) × Read miss rate × Read miss penalty.
#515	**Q515: In a write-through scheme, what are the two sources of write stalls?**
	Answer: Write misses and write buffer stalls.
#516	**Q516: Under what two conditions can write buffer stalls often be safely ignored?**
	Answer: When the system has a reasonably deep write buffer (e.g., four+ words) and a memory that accepts writes much faster than the average write frequency.
#517	**Q517: What simplified formula can be used for Memory-stall clock cycles if read and write penalties are the same and write buffer stalls are negligible?**
	Answer: Memory-stall clock cycles = (Memory accesses/Program) × Miss rate × Miss penalty.
#518	**Q518: What is the formula for memory-stall cycles per instruction?**
	Answer: Memory-stall cycles / Instruction = (Misses / Instruction) × Miss penalty.
#519	**Q519: According to Amdahl's Law, what happens to the fraction of execution time spent on memory stalls if the processor is made faster but the memory system is not?**
	Answer: The fraction of time spent on memory stalls will increase.
#520	**Q520: What is a potential negative consequence of increasing cache size on performance?**
	Answer: It can increase the hit time, which may increase the processor cycle time or add pipeline stages.
#521	**Q521: What metric is used by designers to examine alternative cache designs by considering the time for both hits and misses?**
	Answer: Average memory access time (AMAT).

#522	**Q522: What is the formula for Average Memory Access Time (AMAT)?**
	Answer: AMAT = Time for a hit + (Miss rate × Miss penalty).

#523	**Q523: In a _____ cache, a memory block can be placed in exactly one location.**
	Answer: direct-mapped

#524	**Q524: Term: Fully associative cache**
	Answer: A cache structure in which a block can be placed in any location in the cache.

#525	**Q525: How is a specific block found in a fully associative cache?**
	Answer: All entries in the cache must be searched in parallel using comparators associated with each entry.

#526	**Q526: Term: Set-associative cache**
	Answer: A cache that has a fixed number of locations (at least two) where each block can be placed.

#527	**Q527: Describe the two-step process of locating a memory block in a set-associative cache.**
	Answer: A block is first directly mapped to a specific set, and then all blocks within that set are searched for a match.

#528	**Q528: What formula determines the cache block for a memory block in a direct-mapped cache?**
	Answer: (Block number) modulo (Number of blocks in the cache).

#529	**Q529: What formula determines the set for a memory block in a set-associative cache?**
	Answer: (Block number) modulo (Number of sets in the cache).

#530	**Q530: A direct-mapped cache can be considered a _____-way set-associative cache.**
	Answer: one

#531	**Q531: A fully associative cache with 'm' entries is equivalent to what kind of set-associative cache?**
	Answer: An m-way set-associative cache with only one set.

#532	**Q532: What is the primary advantage of increasing the degree of associativity in a cache?**
	Answer: It usually decreases the miss rate.

#533	**Q533: For a fixed cache size, what is the effect of doubling the associativity on the number of sets and the number of blocks per set?**
	Answer: It halves the number of sets and doubles the number of blocks per set.

#534	**Q534: How does increasing cache associativity by a factor of 2 affect the size of the address's index and tag fields?**
	Answer: It decreases the size of the index by 1 bit and increases the size of the tag by 1 bit.

#535	**Q535: How many index bits are used in a fully associative cache?**
	Answer: Zero; there is no index, as the entire cache is searched.

#536	**Q536: What hardware is required to implement a four-way set-associative cache that is not needed for a direct-mapped cache?**
	Answer: Four parallel comparators and a 4-to-1 multiplexor.

#537	**Q537: Term: Content Addressable Memory (CAM)**
	Answer: A circuit that combines comparison and storage, returning the index of a matching data entry rather than requiring an address.

#538	**Q538: For high degrees of associativity (e.g., eight-way and above), what type of hardware is often used instead of standard SRAMs and comparators?**
	Answer: Content Addressable Memories (CAMs).

#539	**Q539: In an associative cache, what is the most commonly used scheme for choosing which block to replace on a miss?**
	Answer: Least Recently Used (LRU).

#540	**Q540: Term: Least Recently Used (LRU)**
	Answer: A replacement scheme in which the block replaced is the one that has been unused for the longest time.

#541	**Q541: How does increasing associativity affect the total number of tag bits in a cache of a fixed size?**
	Answer: It increases the total number of tag bits because each block's tag field becomes larger.

#542	**Q542: Term: Multilevel cache**
	Answer: A memory hierarchy with multiple levels of caches, rather than just a single cache and main memory.
#543	**Q543: How does a second-level cache improve performance when a miss occurs in the primary cache?**
	Answer: If the data is in the second-level cache, the miss penalty is its access time, which is much lower than accessing main memory.
#544	**Q544: In a system with a primary cache and a secondary cache, what is the formula for Total CPI?**
	Answer: Total CPI = Base CPI + (Primary stalls per instruction) + (Secondary stalls per instruction).
#545	**Q545: In a multilevel cache design, the primary (L1) cache is often designed to minimize _____.**
	Answer: hit time
#546	**Q546: In a multilevel cache design, the secondary (L2) cache is often designed to minimize _____.**
	Answer: miss rate
#547	**Q547: How does a primary cache in a multilevel system typically differ from an optimal single-level cache in terms of size and block size?**
	Answer: It is often smaller and may use a smaller block size.
#548	**Q548: How does a secondary cache in a multilevel system typically differ from an optimal single-level cache in terms of size and associativity?**
	Answer: It is much larger and often uses higher associativity.
#549	**Q549: Why might an algorithm like Quicksort outperform Radix Sort in practice, even if Radix Sort executes fewer instructions for large arrays?**
	Answer: Quicksort can have significantly fewer cache misses, making better use of the memory hierarchy.
#550	**Q550: What is the goal of a blocked algorithm in the context of software optimization for caches?**
	Answer: To maximize accesses to data loaded into the cache before it is replaced, thereby improving temporal locality.
#551	**Q551: The software optimization technique of operating on submatrices instead of entire rows or columns is known as _____.**
	Answer: blocking

#552	**Q552: In matrix multiplication, blocking improves performance by exploiting a combination of spatial locality for one matrix and _____ locality for another.**
	Answer: temporal

#553	**Q553: Term: Global miss rate**
	Answer: The fraction of references that miss in all levels of a multilevel cache.

#554	**Q554: Term: Local miss rate**
	Answer: The fraction of references to one specific level of a cache that miss at that level.

#555	**Q555: Why is the local miss rate of a secondary cache typically much higher than the global miss rate?**
	Answer: Because the primary cache filters out most accesses with good spatial and temporal locality, leaving only harder-to-hit references for the secondary cache.

#556	**Q556: For out-of-order processors, performance evaluation requires simulation because the processor can hide miss latency by _____.**
	Answer: finding other work to do (executing other instructions)

#557	**Q557: Term: Autotuning**
	Answer: An approach where numerical libraries parameterize algorithms and search at runtime to find the best parameter combination for a specific computer's memory hierarchy.

#558	**Q558: Which of the following is generally true about a design with multiple levels of caches?**
	Answer: First-level caches are more concerned about hit time, and second-level caches are more concerned about miss rate.

#559	**Q559: Which cache designer guideline regarding memory bandwidth and block size is generally valid?**
	Answer: The higher the memory bandwidth, the larger the cache block can be without a significant performance penalty.

#560	**Q560: What are the two primary techniques mentioned for improving cache performance?**
	Answer: Reducing the miss rate and reducing the miss penalty.

#561	**Q561: How can the miss rate of a cache be reduced?**
	Answer: By reducing the probability that two distinct memory blocks will contend for the same cache location, for example by using associativity.

#562	**Q562: What technique is used to reduce the miss penalty in a cache hierarchy?**
	Answer: Adding an additional level to the hierarchy, known as multilevel caching.

#563	**Q563: The memory-stall clock cycles in a CPU primarily come from _____.**
	Answer: cache misses

#564	**Q564: For a write-through cache scheme, what are the two main sources of stalls?**
	Answer: Write misses and write buffer stalls.

#565	**Q565: When can write buffer stalls often be safely ignored in performance calculations?**
	Answer: When the system has a reasonably deep write buffer (e.g., four or more words) and memory can accept writes faster than the average write frequency.

#566	**Q566: What is the simplified formula for memory-stall cycles, assuming negligible write buffer stalls and equal read/write penalties?**
	Answer: Memory-stall clock cycles = (Memory accesses/Program) × Miss rate × Miss penalty.

#567	**Q567: How can memory-stall cycles be expressed on a per-instruction basis?**
	Answer: Memory-stall clock cycles = (Instructions/Program) × (Misses/Instruction) × Miss penalty.

#568	**Q568: According to Amdahl's Law, what happens to the fraction of execution time spent on memory stalls if a processor's CPI is reduced but the memory system is not improved?**
	Answer: The fraction of time spent on memory stalls will increase.

#569	**Q569: Term: Average Memory Access Time (AMAT)**
	Answer: Definition: The average time to access memory, considering both hits and misses, calculated as: AMAT = Time for a hit + (Miss rate × Miss penalty).

#570	**Q570: What hardware components make fully associative placement impractical for large caches?**
	Answer: A large number of parallel comparators, one associated with each cache entry.

#571	**Q571: How is the location for a memory block determined in a set-associative cache?**
	Answer: A block is directly mapped to a specific set, and then it can be placed in any of the elements within that set.

#572	**Q572: In a direct-mapped cache, how is the cache block for a memory block determined?**
	Answer: It is determined by the formula: (Block number) modulo (Number of blocks in the cache).

#573	**Q573: In a set-associative cache, how is the set for a memory block determined?**
	Answer: It is determined by the formula: (Block number) modulo (Number of sets in the cache).

#574	**Q574: An m-block fully associative cache can be considered a _____-way set-associative cache.**
	Answer: m

#575	**Q575: What is the main disadvantage of increasing the degree of associativity in a cache?**
	Answer: A potential increase in the hit time due to more complex hardware (comparators and multiplexors).

#576	**Q576: How are all the tags within a selected set of a set-associative cache searched to ensure a fast hit time?**
	Answer: They are searched in parallel.

#577	**Q577: What are the three portions of a memory address used by a set-associative or direct-mapped cache?**
	Answer: The tag, the index, and the block offset.

#578	**Q578: For a fixed cache size, what is the effect of doubling the associativity on the number of sets and the size of the index field?**
	Answer: It halves the number of sets and decreases the size of the index field by 1 bit.

#579	**Q579: For a fixed cache size, what is the effect of doubling the associativity on the size of the tag field?**
	Answer: It increases the size of the tag field by 1 bit.

#580	**Q580: In modern cache design, what level of associativity is typically built using standard SRAMs and comparators versus using CAMs?**
	Answer: Two-way and four-way associativity are often built with SRAMs and comparators, while eight-way and higher associativity are often built using CAMs.

#581	**Q581: What is the most commonly used block replacement scheme for associative caches?**
	Answer: Least Recently Used (LRU).

#582	**Q582: As cache associativity increases, what happens to the difficulty of implementing a true LRU replacement policy?**
	Answer: It becomes harder to implement.

#583	**Q583: What is the purpose of a second-level (L2) cache in a memory hierarchy?**
	Answer: It is accessed on a primary (L1) cache miss to reduce the miss penalty by providing faster access than main memory.

#584	**Q584: What is the formula for the total CPI in a system with two levels of caching?**
	Answer: Total CPI = Base CPI + (Primary misses/instruction × L2 hit penalty) + (Global misses/instruction × Main memory penalty).

#585	**Q585: In a multilevel cache system, what is the primary design focus for the first-level (L1) cache?**
	Answer: Minimizing hit time to enable a shorter clock cycle or fewer pipeline stages.

#586	**Q586: In a multilevel cache system, what is the primary design focus for the second-level (L2) cache?**
	Answer: Minimizing the miss rate to reduce the penalty of long main memory access times.

#587	**Q587: How do the size and block size of a primary cache in a multilevel system typically compare to an optimal single-level cache?**
	Answer: The primary cache is often smaller and may use a smaller block size.

#588	**Q588: How do the size and associativity of a secondary cache in a multilevel system typically compare to an optimal single-level cache?**
	Answer: The secondary cache is much larger and often uses higher associativity.

#589	**Q589: Despite requiring more instructions, why can Quicksort outperform Radix Sort in terms of actual execution time on modern machines?**
	Answer: Quicksort often has significantly fewer cache misses, making better use of the memory hierarchy.

#590	**Q590: What is the primary goal of software optimization via 'blocking'?**
	Answer: To maximize accesses to data loaded into the cache before it's replaced, thereby improving temporal locality and reducing cache misses.

#591	**Q591: Instead of operating on entire rows or columns of an array, blocked algorithms operate on _____.**
	Answer: submatrices or blocks
#592	**Q592: In the context of blocked matrix multiplication, the size of the submatrix is determined by a parameter called the _____.**
	Answer: blocking factor
#593	**Q593: Blocking in matrix multiplication improves performance by exploiting a combination of _____ and _____ locality.**
	Answer: spatial; temporal
#594	**Q594: What is the approximate improvement in the number of memory words accessed when using blocking for matrix multiplication?**
	Answer: The number of memory words accessed is reduced by a factor roughly equal to the BLOCKSIZE.
#595	**Q595: Besides reducing cache misses, what other optimization can blocking be used for?**
	Answer: It can help with register allocation by using a block size small enough to be held in registers, minimizing loads and stores.
#596	**Q596: Why is the local miss rate of a secondary cache typically much higher than its global miss rate?**
	Answer: Because the primary cache filters out accesses with good spatial and temporal locality, leaving only more difficult references for the secondary cache.
#597	**Q597: How is the memory stall calculation for out-of-order processors different from simpler models?**
	Answer: It must account for the 'overlapped miss latency,' which is the portion of the miss penalty that the processor can hide by executing other instructions.
#598	**Q598: As a general guideline, an out-of-order processor can often hide the miss penalty for a miss that hits in the _____ cache, but rarely for a miss to the _____ cache.**
	Answer: L2; L1
#599	**Q599: Which statement is generally true about designing a multilevel cache?**
	Answer: First-level caches are more concerned about hit time, and second-level caches are more concerned about miss rate.

#600	**Q600: How does higher memory bandwidth generally influence the optimal cache block size?**
	Answer: Higher memory bandwidth supports the use of a larger cache block size, as the penalty for fetching a larger block is reduced.

#601	**Q601: For a processor with a 1 ns clock, a miss penalty of 20 cycles, a miss rate of 0.05 misses/instruction, and a hit time of 1 cycle, what is the AMAT?**
	Answer: AMAT = 1 + (0.05 * 20) = 2 clock cycles, or 2 ns.

#602	**Q602: A processor has a CPI of 2 without memory stalls, a 100-cycle miss penalty, an instruction cache miss rate of 2%, and a data cache miss rate of 4%. What are the memory-stall cycles per instruction if 36% of instructions are loads/stores?**
	Answer: Instruction stalls: 1 * 0.02 * 100 = 2.0. Data stalls: 0.36 * 0.04 * 100 = 1.44. Total stall cycles per instruction = 2.0 + 1.44 = 3.44.

#603	**Q603: Consider a cache of 4096 blocks, a 4-word block size, and a 64-bit address. For a two-way set-associative design, how many sets are there?**
	Answer: There are 4096 / 2 = 2048 sets.

#604	**Q604: Consider a cache of 4096 blocks, a 4-word block size, and a 64-bit address. For a two-way set-associative design, how many bits are in the tag?**
	Answer: The address has 60 bits for tag+index. The index needs log2(2048) = 11 bits. So, the tag is 60 - 11 = 49 bits.

Chapter 5.8 - Memory Hierarchy Framework & Block Placement

#605	**Q605: What are the four fundamental questions that provide a common framework for understanding any memory hierarchy?**
	Answer: 1. Where can a block be placed? 2. How is a block found? 3. Which block should be replaced on a miss? 4. How are writes handled?

#606	**Q606: In a memory hierarchy, what is the term for a block placement scheme where a block can be placed in exactly one location?**
	Answer: Direct mapped.

#607	**Q607: In a _____ placement scheme, a block can be placed in a restricted set of places in the cache.**
	Answer: set-associative

#608	**Q608: What placement scheme allows a block to be placed anywhere in the upper level of the memory hierarchy?**
	Answer: Fully associative.

#609	**Q609: What is the primary advantage of increasing the degree of associativity in a memory hierarchy?**
	Answer: It usually decreases the miss rate by reducing misses that compete for the same location.

#610	**Q610: According to the source, what is the typical percentage reduction in miss rate when moving from a direct-mapped to a two-way set-associative cache?**
	Answer: Between 20% and 30%.

#611	**Q611: What are the two potential disadvantages of increasing associativity in a cache?**
	Answer: Increased cost and slower access time.

#612	**Q612: Why does the absolute improvement in miss rate from associativity shrink as cache sizes increase?**
	Answer: Because the overall miss rate of a larger cache is already lower, leaving less opportunity for improvement.

#613	**Q613: How is a block found in a direct-mapped cache?**
	Answer: By using an index, which requires only one comparison.

#614	**Q614: How is a block located in a set-associative cache?**
	Answer: By first indexing the set and then searching among the elements within that set.

#615	**Q615: What type of search is required to find a block in a fully associative cache?**
	Answer: A full search of all cache entries.

#616	**Q616: What data structure is used to find a block in a virtual memory system?**
	Answer: A separate lookup table, specifically the page table.

#617	**Q617: Why are fully associative caches typically prohibitive except for small sizes?**
	Answer: The cost of the comparators required for a full search becomes overwhelming for large caches.

#618	**Q618: What block placement scheme is almost always used for virtual memory systems?**
	Answer: Fully associative placement.

#619	**Q619: What are the three motivating factors for using fully associative placement in virtual memory systems?**
	Answer: Misses are very expensive, it allows for sophisticated software replacement schemes, and the map can be easily indexed.

#620	**Q620: What is the primary advantage of a direct-mapped cache design?**
	Answer: Faster access time and simplicity, as finding the block does not depend on a comparison.

#621	**Q621: In an associative cache, which replacement strategy involves selecting a candidate block at random?**
	Answer: Random replacement.

#622	**Q622: In an associative cache, what does the Least Recently Used (LRU) replacement strategy dictate?**
	Answer: The block that has been unused for the longest time should be replaced.

#623	**Q623: Why is a true LRU replacement policy costly to implement for caches with high associativity (more than four-way)?**
	Answer: Because tracking the detailed usage information for many blocks is expensive in hardware.

#624	**Q624: For a two-way set-associative cache, how does the miss rate of random replacement compare to LRU replacement?**
	Answer: The miss rate for random replacement is about 1.1 times higher than for LRU.

#625	**Q625: What type of replacement algorithm is always approximated in virtual memory, and why?**
	Answer: Some form of LRU is always approximated because the cost of a miss is enormous, making even a tiny reduction in miss rate important.

#626	**Q626: What is a 'write-through' policy in a memory hierarchy?**
	Answer: Information is written to both the block in the cache and the block in the lower level of the memory hierarchy simultaneously.

#627	**Q627: What is a 'write-back' policy in a memory hierarchy?**
	Answer: Information is written only to the block in the cache, and the modified block is written to the lower level only when it is replaced.
#628	**Q628: What are the three key advantages of a write-back policy?**
	Answer: Processor can write at cache speed, multiple writes to a block require only one lower-level write, and it allows for high-bandwidth transfers.
#629	**Q629: What are the two main advantages of a write-through policy?**
	Answer: Misses are simpler and cheaper, and the policy is easier to implement.
#630	**Q630: Why is a write-back policy considered the only practical option for virtual memory systems?**
	Answer: Because of the very long latency of a write to the lower level of the hierarchy (e.g., a disk).
#631	**Q631: Which write policy is typically used by lowest-level caches today, and why?**
	Answer: Write-back, because the rate of processor writes generally exceeds the rate at which the memory system can process them.
#632	**Q632: What is the 'three Cs' model used for?**
	Answer: It provides an intuitive model for classifying cache misses into one of three categories to understand memory hierarchy behavior.
#633	**Q633: What are the three categories of misses in the 'three Cs' model?**
	Answer: Compulsory misses, capacity misses, and conflict misses.
#634	**Q634: Define: Compulsory Miss**
	Answer: A cache miss caused by the first access to a block that has never been in the cache; also called a cold-start miss.
#635	**Q635: Define: Capacity Miss**
	Answer: A cache miss that occurs because the cache cannot contain all the blocks needed during a program's execution, even with full associativity.

#636	**Q636: Define: Conflict Miss**
	Answer: A miss in a direct-mapped or set-associative cache that occurs when multiple blocks compete for the same set and would be eliminated in a fully associative cache of the same size.

#637	**Q637: What is the primary way to reduce the number of compulsory misses?**
	Answer: Increase the block size.

#638	**Q638: What is a straightforward way to reduce capacity misses?**
	Answer: Enlarge the cache size.

#639	**Q639: How can conflict misses be reduced in a cache design?**
	Answer: By increasing the associativity.

#640	**Q640: What is the potential negative performance effect of increasing cache size to reduce capacity misses?**
	Answer: It may increase the cache access time.

#641	**Q641: What is the potential negative performance effect of increasing associativity to reduce conflict misses?**
	Answer: It may increase the cache access time.

#642	**Q642: What are the potential negative performance effects of increasing the block size to reduce compulsory misses?**
	Answer: It increases the miss penalty, and a very large block size could increase the overall miss rate.

#643	**Q643: According to the 'Check Yourself' section, is it true that there is no way to reduce compulsory misses?**
	Answer: No, this is false; increasing the block size can reduce compulsory misses.

#644	**Q644: According to the 'Check Yourself' section, is it true that fully associative caches have no conflict misses?**
	Answer: Yes, this is true by definition, as conflict misses are those eliminated by a fully associative cache.

#645	**Q645: As cache sizes grow, the relative improvement from associativity _____, and the absolute improvement _____**
	Answer: increases only slightly; shrinks significantly

#646	**Q646: Why do smaller caches obtain a significantly larger absolute benefit from associativity?**
	Answer: Because the base miss rate of a small cache is larger, providing more opportunity for improvement.

#647	**Q647: The method of finding a block in a memory hierarchy is dictated by the _____ scheme.**
	Answer: block placement

#648	**Q648: What feature of on-chip L2 caches enables them to have much higher associativity compared to off-chip caches?**
	Answer: Their hit times are not as critical, and designers are not constrained by standard SRAM chip building blocks.

#649	**Q649: In a direct-mapped cache, how many candidate blocks are there for replacement on a miss?**
	Answer: Only one.

#650	**Q650: In virtual memory systems, what hardware feature is often provided to help the OS approximate an LRU replacement policy?**
	Answer: Reference bits or equivalent functionality to track page usage.

#651	**Q651: In the 'three Cs' model, the total miss rate of a direct-mapped cache is the sum of compulsory, capacity, and conflict misses from one-way, two-way, and _____ associativity.**
	Answer: four-way

#652	**Q652: What are the four fundamental questions that provide a common framework for understanding any level of a memory hierarchy?**
	Answer: 1. Where can a block be placed? 2. How is a block found? 3. Which block should be replaced on a miss? 4. What happens on a write?

#653	**Q653: In a direct-mapped placement scheme, how many sets are there and how many blocks are in each set?**
	Answer: There are as many sets as blocks in the cache, with exactly one block per set.

#654	**Q654: In a fully associative placement scheme, how many sets are there and how many blocks are in each set?**
	Answer: There is only one set, which contains all the blocks in the cache.

#655	**Q655: What is the primary advantage of increasing the degree of associativity in a cache?**
	Answer: It usually decreases the miss rate by reducing misses that compete for the same location (conflict misses).

#656	**Q656: What is the typical percentage reduction in miss rate when moving from a direct-mapped (one-way) to a two-way set-associative cache?**
	Answer: The miss rate is typically reduced by 20% to 30%.

#657	**Q657: What are the two potential disadvantages of increasing a cache's associativity?**
	Answer: Increased cost and slower access time.

#658	**Q658: As cache sizes grow, the _____ improvement in the miss rate from associativity shrinks significantly.**
	Answer: absolute

#659	**Q659: How is a block found in a set-associative cache?**
	Answer: By indexing to find the correct set and then searching among the elements within that set.

#660	**Q660: How is a block found in a fully associative cache?**
	Answer: By searching through all entries in the entire cache.

#661	**Q661: What mechanism is used to find a block (page) in virtual memory systems?**
	Answer: A separate lookup table, known as the page table.

#662	**Q662: What is the first reason virtual memory systems almost always use fully associative placement?**
	Answer: Full associativity is beneficial because memory misses are extremely expensive in terms of performance.

#663	**Q663: What is the second reason virtual memory systems use fully associative placement?**
	Answer: It allows software to use sophisticated replacement schemes to reduce the miss rate.

#664	**Q664: What is the third reason virtual memory systems use fully associative placement?**
	Answer: The full map (page table) can be easily indexed with no extra hardware or searching required.

#665	**Q665: What is the primary advantage of a direct-mapped cache that might lead a designer to choose it?**
	Answer: Its advantage in access time and simplicity, as finding a block does not depend on a comparison.

#666	**Q666: For which types of caches must a block replacement policy be decided upon when a miss occurs?**
	Answer: Set-associative and fully associative caches.

#667	**Q667: What are the two primary strategies for block replacement in associative caches?**
	Answer: Random and Least Recently Used (LRU).

#668	**Q668: Define the Least Recently Used (LRU) replacement policy.**
	Answer: The block that is replaced is the one that has been unused for the longest time.

#669	**Q669: Why is a true LRU policy too costly to implement for caches with a high degree of associativity (more than four-way)?**
	Answer: Because tracking the detailed usage information required for true LRU is expensive in terms of hardware.

#670	**Q670: For a two-way set-associative cache, how does the miss rate of a random replacement policy compare to an LRU policy?**
	Answer: The miss rate for random replacement is about 1.1 times higher than for LRU replacement.

#671	**Q671: Why is some form of LRU always approximated for replacement in virtual memory?**
	Answer: Because the cost of a miss is enormous, so even a tiny reduction in the miss rate is very important.

#672	**Q672: The _____ write policy involves writing information to both the block in the cache and the block in the lower level of the memory hierarchy.**
	Answer: write-through

#673	**Q673: The _____ write policy involves writing information only to the block in the cache, and to the lower level only when that block is replaced.**
	Answer: write-back

#674	**Q674: What is an advantage of write-back concerning processor write speed?**
	Answer: Individual words can be written at the speed of the cache, not the slower main memory.

#675	**Q675: What is an advantage of write-back concerning multiple writes to the same block?**
	Answer: Multiple writes within a single block require only one write to the lower-level memory when the block is finally replaced.

#676	**Q676: What is an advantage of write-through concerning cache misses?**
	Answer: Misses are simpler and cheaper because they never require a modified block to be written back to the lower level.

#677	**Q677: Why is write-back the only practical write policy for virtual memory systems?**
	Answer: Because of the extremely long latency of a write to the lower level of the hierarchy (e.g., a disk).

#678	**Q678: Which write policy do lowest-level caches typically use today and why?**
	Answer: Write-back, because the rate of processor writes generally exceeds the rate at which the memory system can process them.

#679	**Q679: What are the "Three Cs" used to classify all cache misses?**
	Answer: Compulsory misses, capacity misses, and conflict misses.

#680	**Q680: Term: Compulsory miss**
	Answer: A cache miss caused by the first access to a block that has never been in the cache; also called a cold-start miss.

#681	**Q681: Term: Capacity miss**
	Answer: A cache miss that occurs because the cache cannot contain all the blocks needed by the program, even with full associativity.

#682	**Q682: Term: Conflict miss**
	Answer: A cache miss in a direct-mapped or set-associative cache that occurs when multiple blocks compete for the same set.

#683	**Q683: A conflict miss would be eliminated if the cache were _____ of the same size.**
	Answer: fully associative

#684	**Q684: What is the most direct way to reduce the number of conflict misses?**
	Answer: By increasing the associativity of the cache.

#685	**Q685: What is the most direct way to reduce the number of capacity misses?**
	Answer: By enlarging the total size of the cache.

#686	**Q686: What is a potential negative performance impact of increasing the cache size?**
	Answer: It may increase the cache access time.

#687	**Q687: What is a potential negative performance impact of increasing the block size?**
	Answer: It increases the miss penalty (the time to fetch the larger block).

#688	**Q688: According to the 'Check Yourself' section, do fully associative caches have conflict misses?**
	Answer: No, by definition, fully associative caches have no conflict misses.

#689	**Q689: As cache associativity increases from four-way to eight-way, what happens to the improvement in miss rate?**
	Answer: There is very little improvement in the miss rate compared to the gain from one-way to two-way.

#690	**Q690: What is a key trade-off that makes memory hierarchy design challenging?**
	Answer: Every change that can improve the miss rate (like increasing size or associativity) can also negatively affect overall performance (like increasing access time).

#691	**Q691: In the "Three Cs" model graph (Figure 5.36), which component of the miss rate is so small it is often not visible?**
	Answer: The compulsory miss component.

#692	**Q692: Why have first-level caches been growing slowly, if at all, compared to second-level caches?**
	Answer: To avoid increasing access time, which could lower overall performance, as first-level cache access time is often critical to processor cycle time.

#693	**Q693: In a set-associative cache, how many blocks are candidates for replacement on a miss?**
	Answer: All the blocks within the specific set where the miss occurred.

#694	**Q694: How is the location method for a block in a memory hierarchy determined?**
	Answer: It is dictated by the block placement scheme (direct-mapped, set-associative, or fully associative).